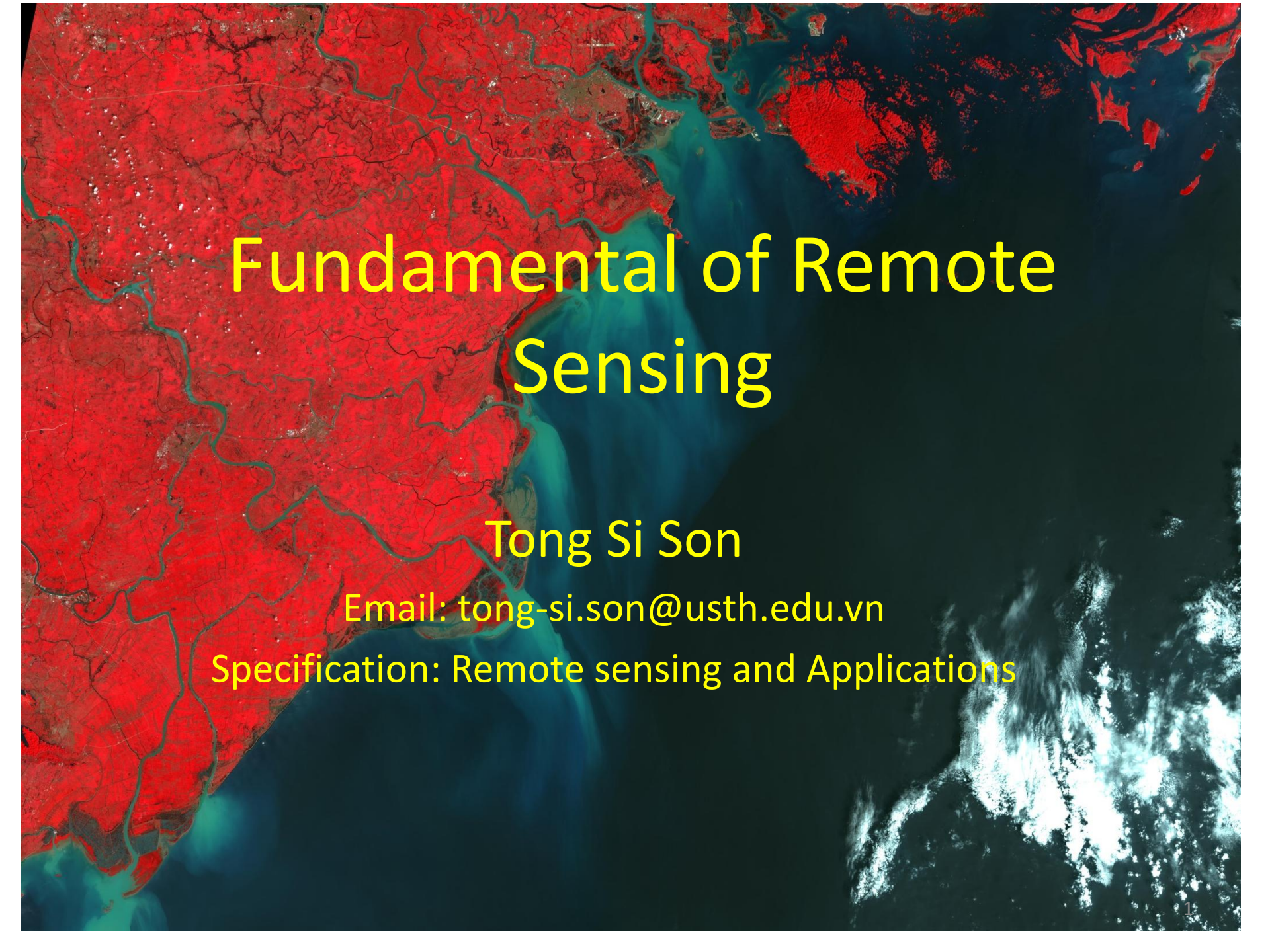


A satellite image of a coastal region. The land is colored red, showing a complex network of rivers and waterways. The water is dark blue. In the bottom right corner, there is a white, cloud-like feature. The title 'Fundamental of Remote Sensing' is overlaid in yellow text.

Fundamental of Remote Sensing

Tong Si Son

Email: tong-si.son@usth.edu.vn

Specification: Remote sensing and Applications

Structure Of The Course

- **Time**

36 hours: Lecturer: 21 h, Practice: 15 h

- **Contents**

1. Overview of Remote sensing

- 2. Physical basis of Remote sensing**

3. Remote sensing system

4. Satellite image characteristics

5. Satellite image pre-processing

6. Satellite image classification and accuracy assessment

Section 2. Physical basis of Remote sensing

(Material by Prof. Gabriëlle De Lannoy, KU Leuven)

2.1 Electromagnetic Radiation

2.2 Interactions With The Atmosphere

2.3 Interactions At The Earth's Surface

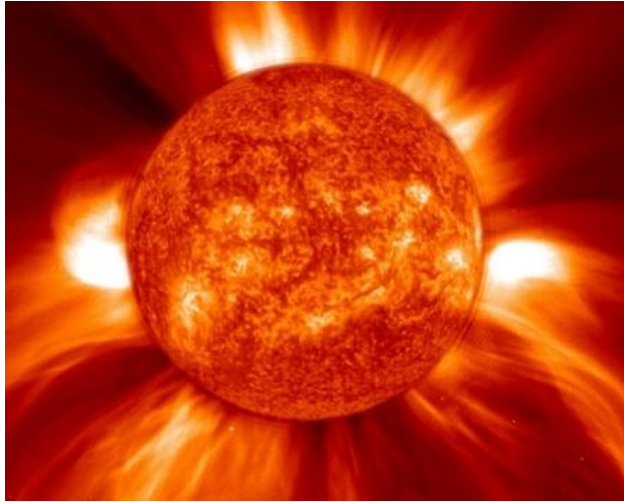
2.4 Interaction At The Sensor

What make the color of landscape?



The certain spectral wavelength reflected from an element more than the other wavelengths create the color
Remote sensing measure radiation to provide information

Sources of electromagnetic radiation



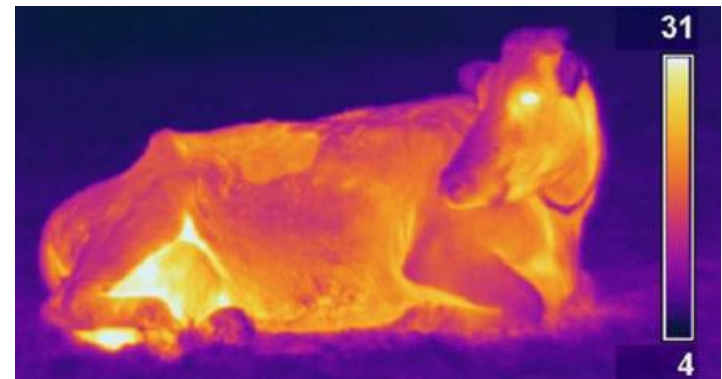
Sun: 6000k



Earth: 300k



Antena: long wave radiation (Radar)

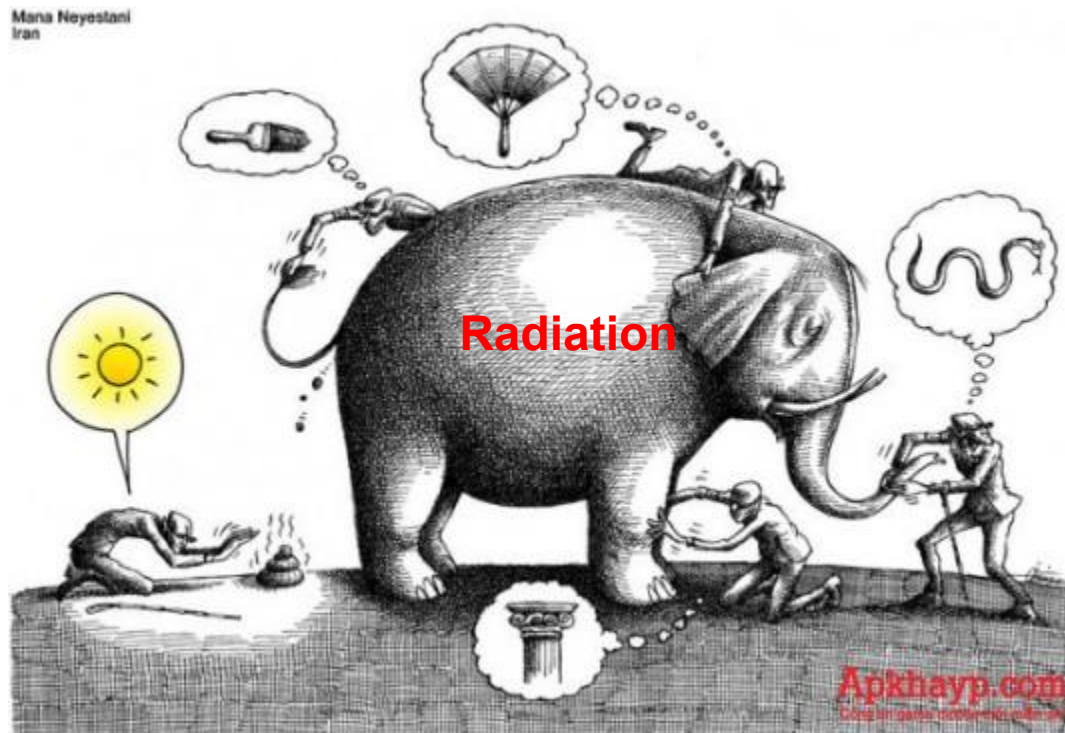


Animal: 310k

Remote sensing

Most of the time, not direct measurements of what we want to observe
Satellites observe radiation, NOT vegetation, soil moisture or wind speed...

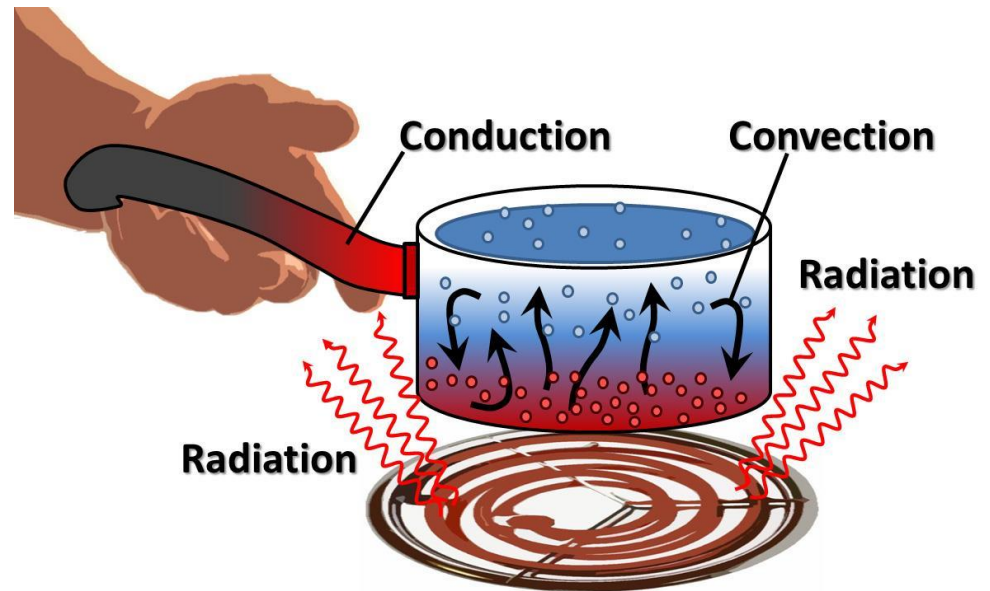
Information to be extracted from the radiation emitted, reflected, or scattered
by the surface and the atmosphere (inversion problem)



2.1 Electromagnetic radiation

Energy transfer

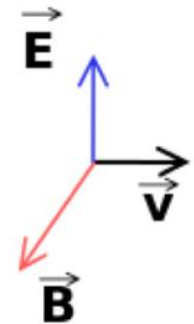
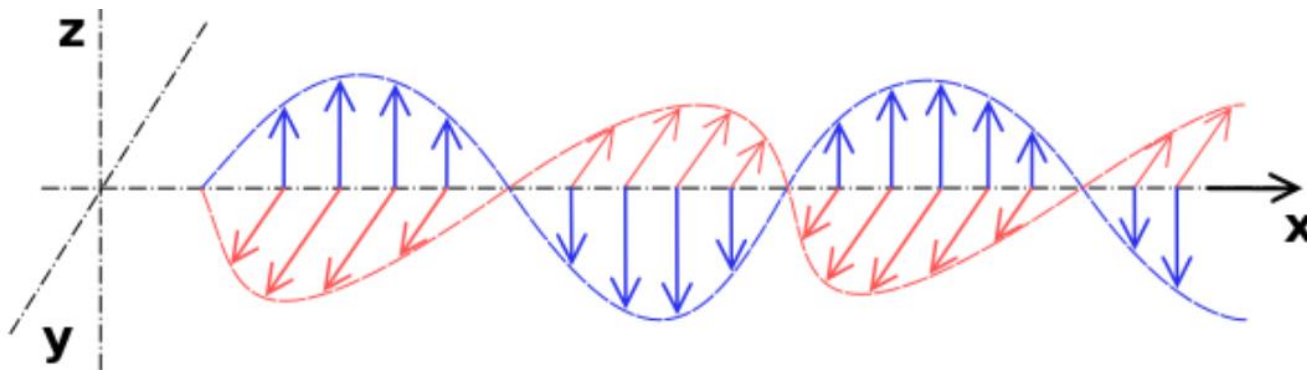
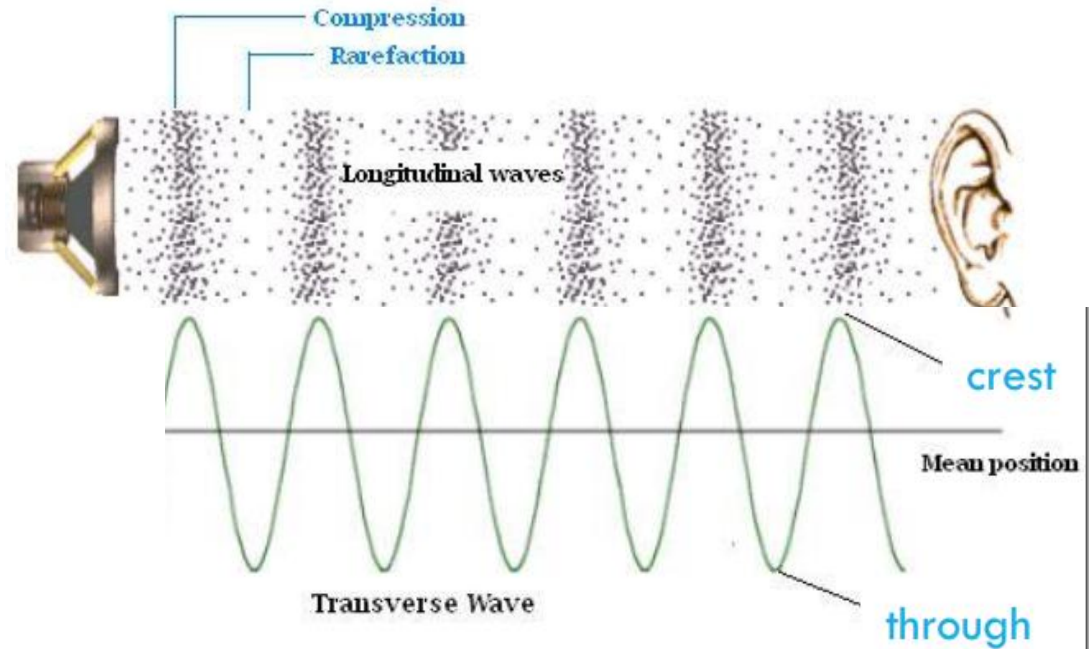
- **Conduction:** transfer internal kinetic and potential (=internal) energy by microscopic diffusion and collision of particles.
- **Convection:** collective movement of molecules within a fluid (liquid, gas) by diffusion and advection
- **Radiation:** only energy transfer that can take place in vacuum (=no medium), exhibits both particle and wave properties



RADIATION - WAVES

Electromagnetic radiation

- Synchronized oscillations of electric and magnetic fields
- Speed of light through vacuum C
- Maxwell equations: electric and magnetic fields are perpendicular
- Transverse waves



RADIATION - WAVES

The amplitude determines the intensity of light.

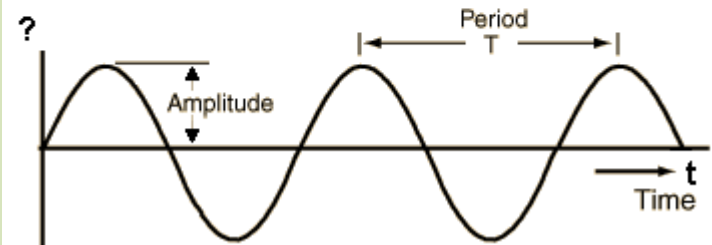
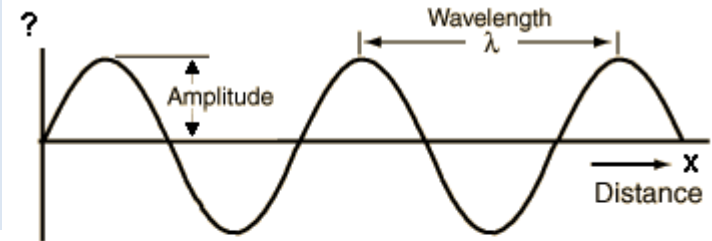
Periodic properties

Wavelength, λ	
kilometer [km]	10^3 m
meter [m]	1 m
centimeter [cm]	10^{-2} m
millimeter [mm]	10^{-3} m
micrometer [μ m]	10^{-6} m
nanometer [nm]	10^{-9} m
Angstrom [$\text{\AA}$]	10^{-10} m
Frequency, f	
gigahertz (GHz)	10^9 Hz
megahertz (MHz)	10^6 Hz
kilohertz (kHz)	10^3 Hz
hertz (Hz)	1 Hz = 1 / s

The wavenumber is the number of wavelengths per unit distance:
 $\nu = 1 / \lambda$

The wavelength λ and period T are related by the speed of light:
 $c = \lambda / T$
 $c = 300,000$ km/s

The frequency is the inverse of the period: $f = 1 / T$



RADIATION - WAVES

The wavelength λ and period T are related by the speed of light:

$$c = \lambda / T$$

$$f = 1 / T$$

$$c = f \lambda$$

$$c = 300,000 \text{ km/s}$$

Intensity, or irradiance = power/area [W/m^2]
(depend on Amplitude, Frequency,
 $1/\text{Wavelength}$)

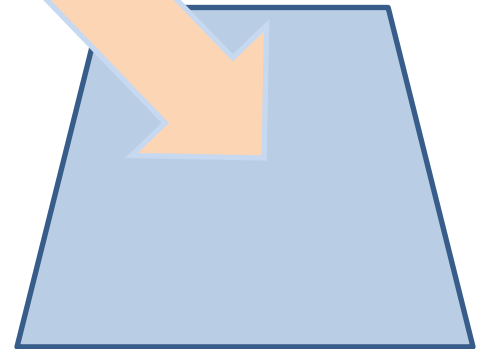
Niels Bohr and Max Planck

$$Q = h c / \lambda = h f$$

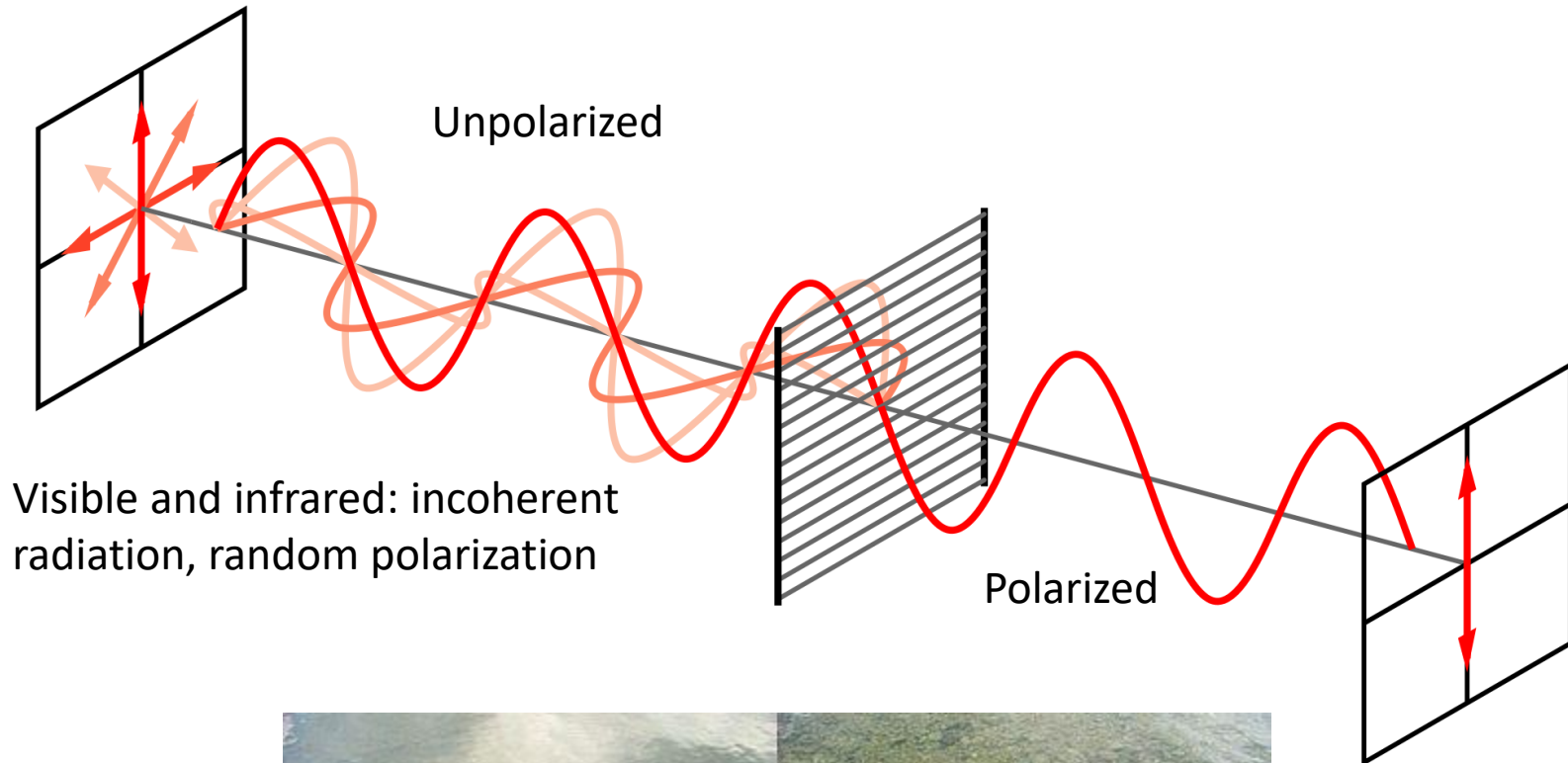
Q ; the energy of a quantum (J)

h ; the Planck constant = $6.626 \times 10^{-34} \text{ J s}^{-1}$

λ ; wave length.



RADIATION - WAVES



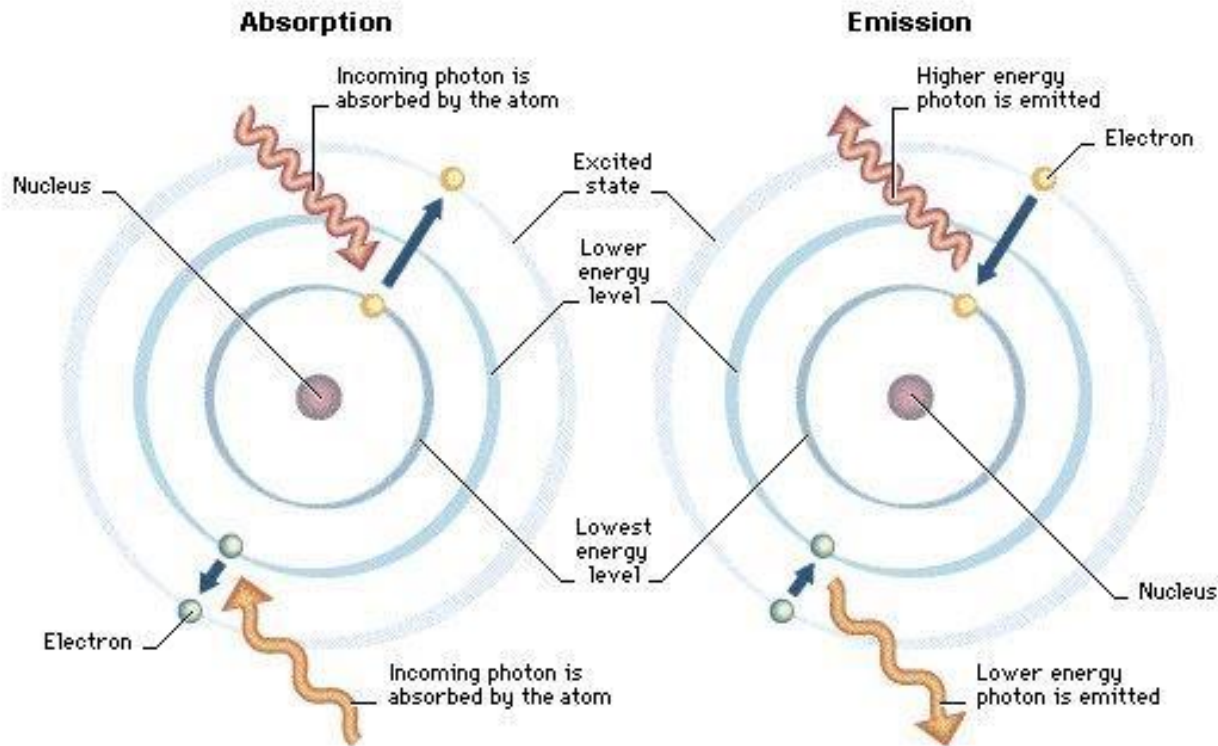
RADIATION - PHOTONS

- Using a photomultiplier and oscilloscope, light can be demonstrated as particles
- Einstein, Bohr
- Photons generated by electron motion from three sources:

Electron transition: fall from high to low orbital energy level

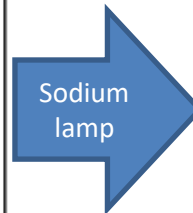
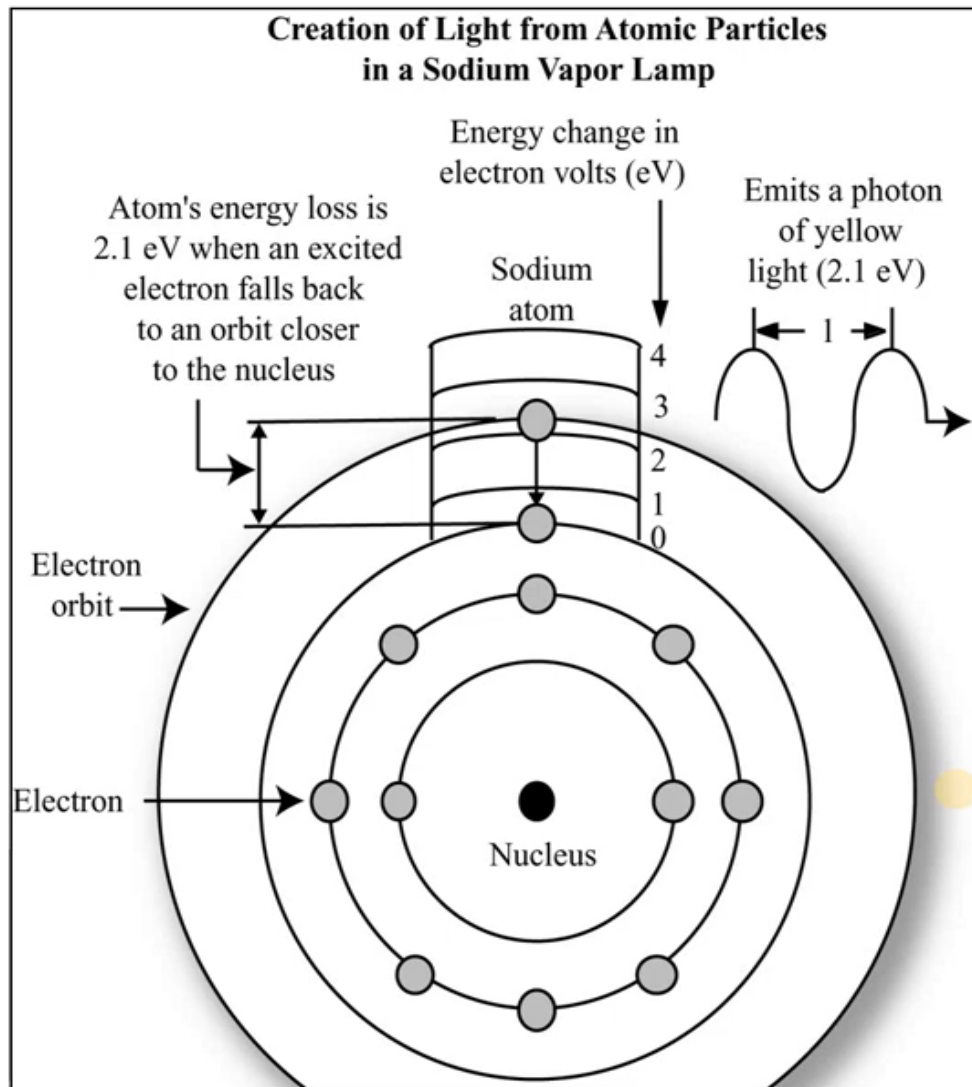
Molecular vibration: atoms + electrons vibration as part of a molecule

Molecular rotation: molecules spinning along an axis



RADIATION - PHOTONS

Electron transitions about 10 nanoseconds (10^{-8} seconds)



Street light

$Q = hf = \frac{hc}{\lambda} = 2.1 \text{ eV} \rightarrow$ the wavelength? Color of light?

with

Q = amount of energy per photon [J]

h = Planck's constant = $6.63 \times 10^{-34} \text{ J.s} = 4.1356 \times 10^{-15} \text{ eV.s}$

f = frequency of the radiation [1/s]

What is the color of light?

Example of electric charges of Sodium vapor lamp

RADIATION - PHOTONS

- Radiation emitted by blackbodies
- Absorption and emission of spectral lines by atoms and molecules

Blackbody

- Blackbody: completely absorbs all light, does not show any reflection
- Blackbody radiation, **total** emitted radiation [W/m²] : $M = \sigma.T^4$
with T temperature [K]

σ : Stefan-Boltzmann constant $5.67 \times 10^{-8} \text{ W/m}^2/\text{K}^4$

M = integration over all wavelengths in the Planck spectrum (from radio waves to gamma rays)

Emitted radiation (W/m²) in a specific wavelength

$$M_{\lambda} = \frac{2\pi hc^2}{\lambda^5 (e^{hc/\lambda kT} - 1)}$$

k Boltzmann constant = $1.380649 \times 10^{-23} \text{ (J/T_Kenvil)}$

h Planck's constant = $6.63 \times 10^{-34} \text{ J.s} = 4.1356 \times 10^{-15} \text{ eV.s}$

c Speed of light in vacuum = $300,000 \text{ km/s}$

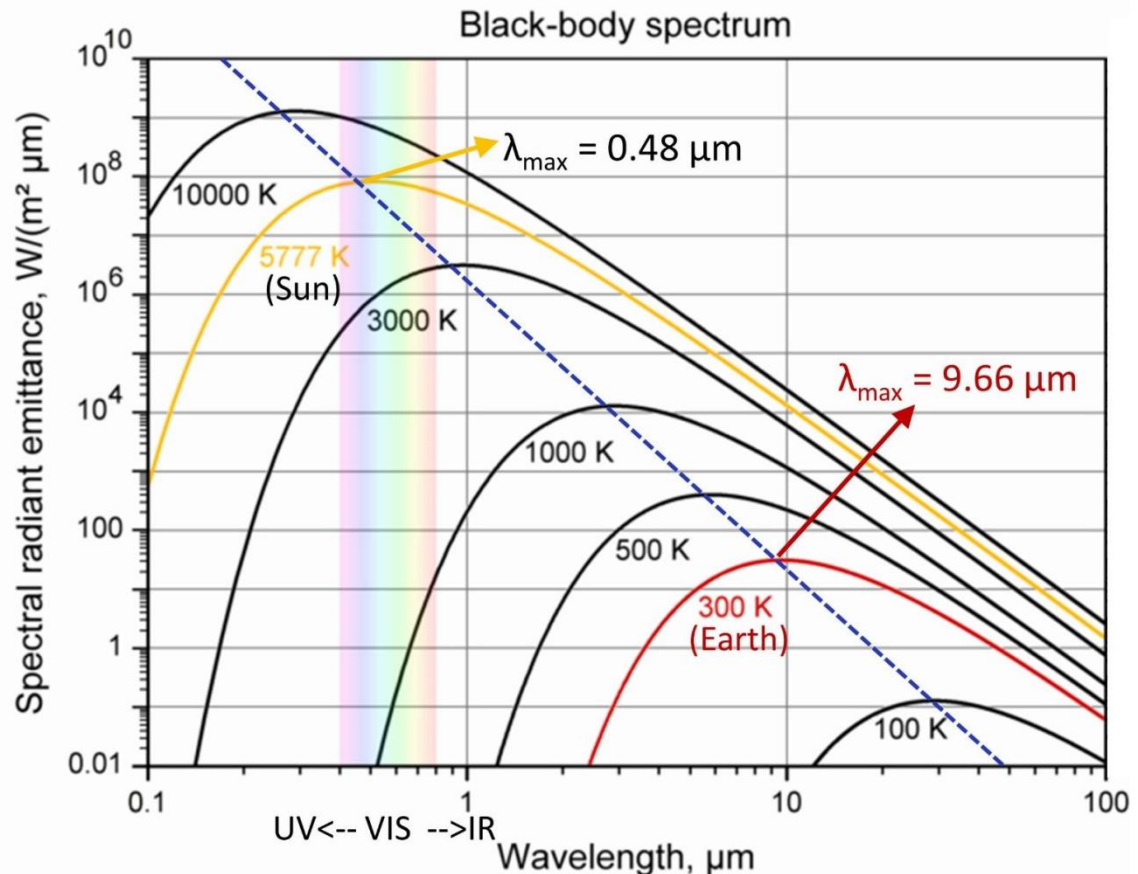
RADIATION - PHOTONS

- Emittance = exitance $M_\lambda(\lambda)$

Spectral emittance = radiant flux (W)/area (m²)/ wavelength(μm) =
radiant exitance (W/m²)/wavelength (μm)

Wien's displacement law:
 $\lambda_{\max} \cdot T = b$
Constant $b = 2,898 \mu\text{m}\cdot\text{K}$

Higher temperature
Larger total emitted radiation
Shorter dominant wavelength



What is the temperature of the street light in the slide 13?

RADIATION - PHOTONS

- **Quantum theory**

Photons describe the amount of energy Q transmitted (and measured by the remote sensor)

$$Q = h f = h.c/\lambda$$

with

Q = amount of energy per photon [J]

h = Planck's constant = $6.63 \cdot 10^{-34}$ J.s = $4.1356 \cdot 10^{-15}$ eV.s

f = frequency of the radiation [1/s]

The energy Q of a photon is inversely proportional to its wavelength: the longer the wavelength involved, the lower its energy content (important to RS: more difficult to detect longer-wavelength radiation !)

RADIATION - PHOTONS

- Energy Q [J]

>> Power = $dQ/d\text{time} = \Phi$ [W]

>> Flux density, irradiance M, E [W/m^2]

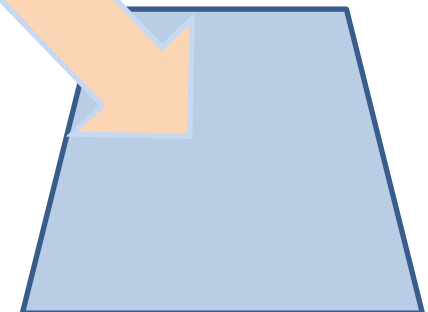
- By wavelength

>> $\Phi_\lambda(\lambda)$ [W/m]

>> $E_\lambda(\lambda), M_\lambda(\lambda)$ [W/m^2]

Irradiance = power/area [W/m^2]
= $dQ/d\text{time}/\text{area}$ [$\text{J}/\text{s}/\text{m}^2$]

*(depend on Amplitude, Frequency,
1/Wavelength)*

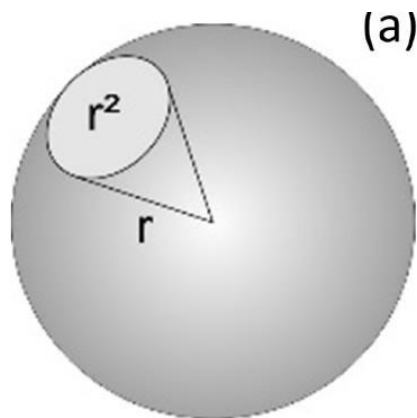


RADIATION

Quantity	Definition
Radiant flux or radiant power Φ [W]	total power of radiation emitted by a source, without any information on the spectral distribution or the directional distribution of the output
Radiant exitance M [W/m ²]	radiant power per area, emitted or reflected from a certain location on a surface
Irradiance E [W/m ²]	radiant power impinging upon a surface per unit area
Spectral radiant power $\Phi_\lambda(\lambda)$, spectral radiant exitance $M_\lambda(\lambda)$, and spectral radiant irradiance $E_\lambda(\lambda)$	See above, but quantified per wavelength

RADIATION

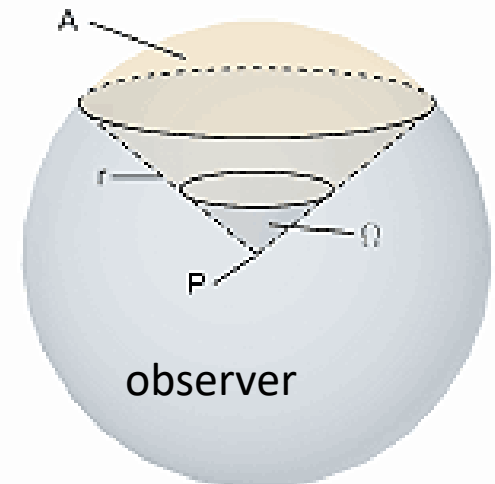
Quantity	Definition
Solid angle Ω [sr], or $[-] = A/r^2$	quantifies a part of an observer's visual field
Radiant intensity I [W/sr]	radiant power of a source emitted in a certain direction
Radiance L [W/m ² /sr]	radiation emitted or reflected from a certain location on an emitting or reflecting surface in a particular direction (radiant exitance in a particular direction)
Spectral radiant intensity $I_\lambda(\lambda)$ [W/m/sr], spectral radiant radiance $L_\lambda(\lambda)$ [W/m ² /sr]	See above, but quantified per wavelength



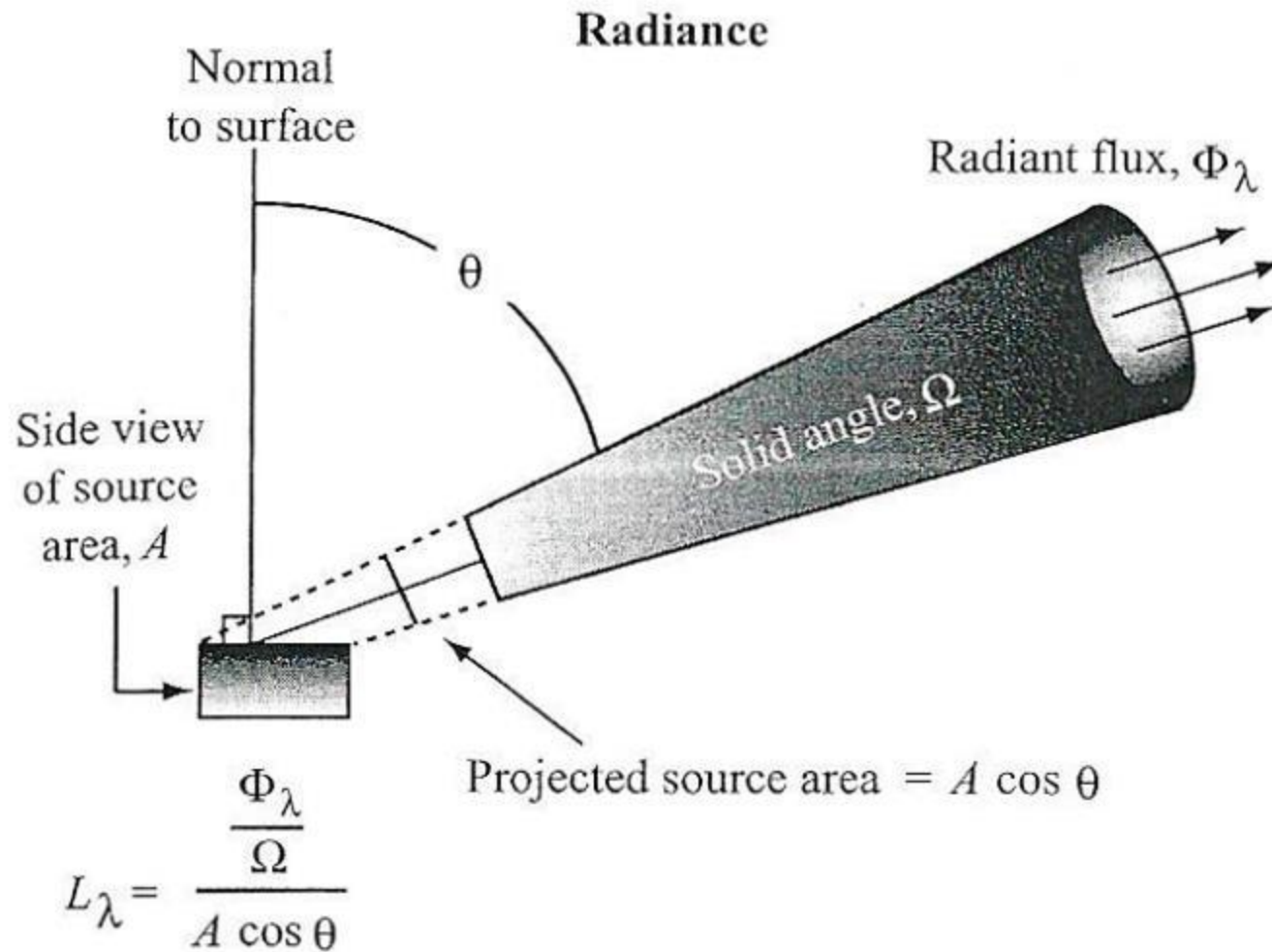
(a)

Solid angle : $\Omega = S_\Omega / r^2$
 where S_Ω is the spherical surface area supported by the solid angle Ω in a sphere of radius r .

If $(S_\Omega = r^2) \Rightarrow \Omega = 1$ steradian (1 sr)

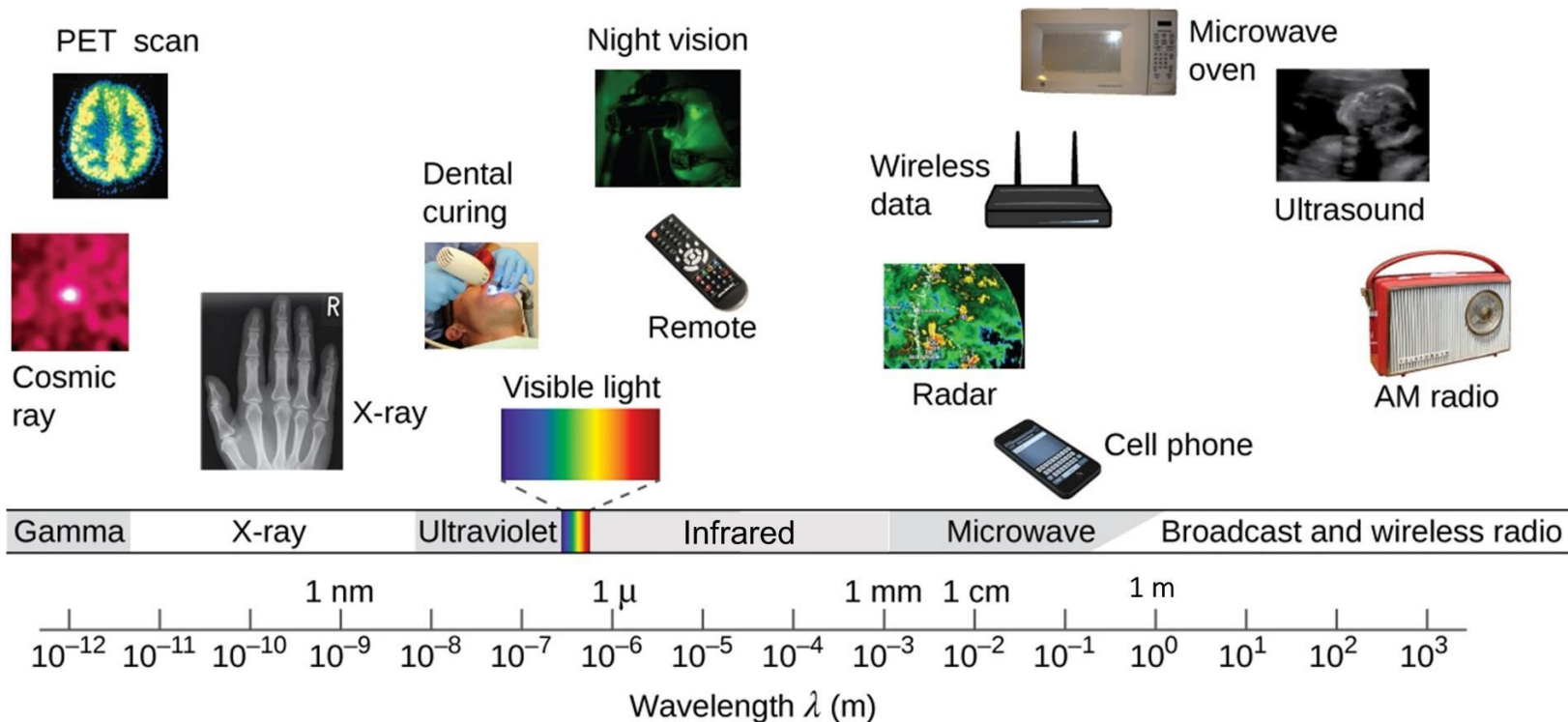
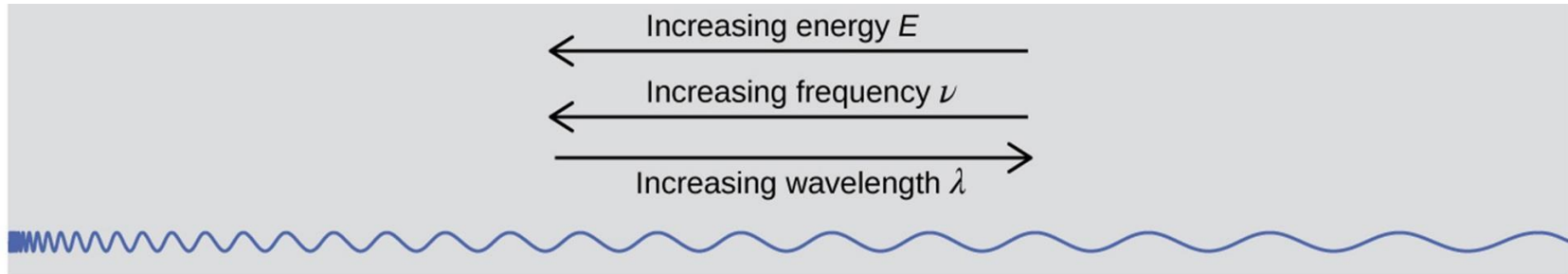


RADIANCE L [$\text{W}/\text{M}^2/\text{SR}$]



ELECTROMAGNETIC RADIATION

EM spectrum = range of all types of EM radiation



Bands,
channels or
regions in
the
spectrum

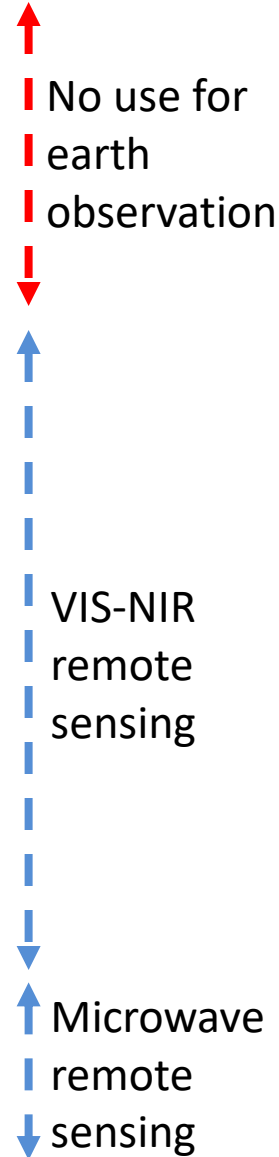
$$c = \lambda \cdot f$$

$$f = c / \lambda$$

$$Q = h \cdot f$$

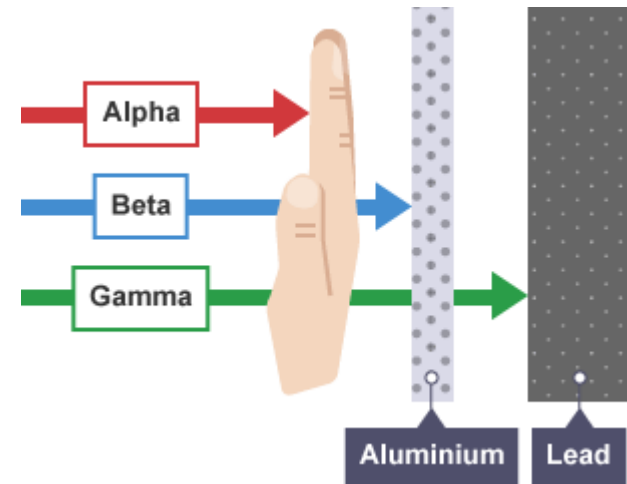
Therefore: $Q = h \cdot c / \lambda$

Wave	length [m]	frequency [Hz]
Cosmic Rays	10^{-15} to 10^{-20}	10^{20} to 10^{22}
Gamma Rays	10^{-11} to 10^{-15}	$3 \cdot 10^{19}$ to $3 \cdot 10^{20}$
X-Rays	10^{-9} to 10^{-11}	$3 \cdot 10^{17}$ to $3 \cdot 10^{19}$
Ultraviolet	$0.4 \cdot 10^{-6}$ to 10^{-9}	$7.5 \cdot 10^{14}$ to $3 \cdot 10^{17}$
Ultraviolet, sw	$0.254 \cdot 10^{-6}$	$11.82 \cdot 10^{14}$
Ultraviolet, lw	$0.366 \cdot 10^{-6}$	$8.19 \cdot 10^{14}$
Violet ('limit')	$0.40 \cdot 10^{-6}$	$7.50 \cdot 10^{14}$
Blue	$0.45 \cdot 10^{-6}$	$6.66 \cdot 10^{14}$
Green	$0.50 \cdot 10^{-6}$	$6.00 \cdot 10^{14}$
Green('limit')	$0.55 \cdot 10^{-6}$	$5.45 \cdot 10^{14}$
Yellow	$0.58 \cdot 10^{-6}$	$5.17 \cdot 10^{14}$
Orange	$0.60 \cdot 10^{-6}$	$5.00 \cdot 10^{14}$
Red	$0.65 \cdot 10^{-6}$	$4.62 \cdot 10^{14}$
Red ('limit')	$0.7 \cdot 10^{-6}$	$4.29 \cdot 10^{14}$
Infrared, near	$1.0 \cdot 10^{-6}$	$3.00 \cdot 10^{14}$
Infrared, far	$30.0 \cdot 10^{-6}$	$0.10 \cdot 10^{14}$
Infrared	10^{-3} to $0.7 \cdot 10^{-6}$	$3 \cdot 10^{12}$ to $4.3 \cdot 10^{14}$
Microwave	0.5 to 10^{-3}	10^8 to $3 \cdot 10^{12}$
Radiowave	>0.5	10^8 to 1

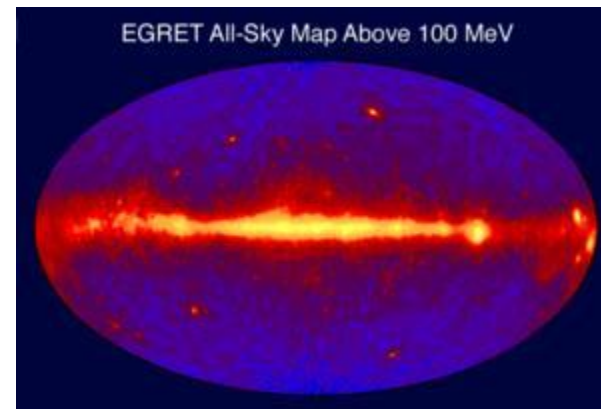
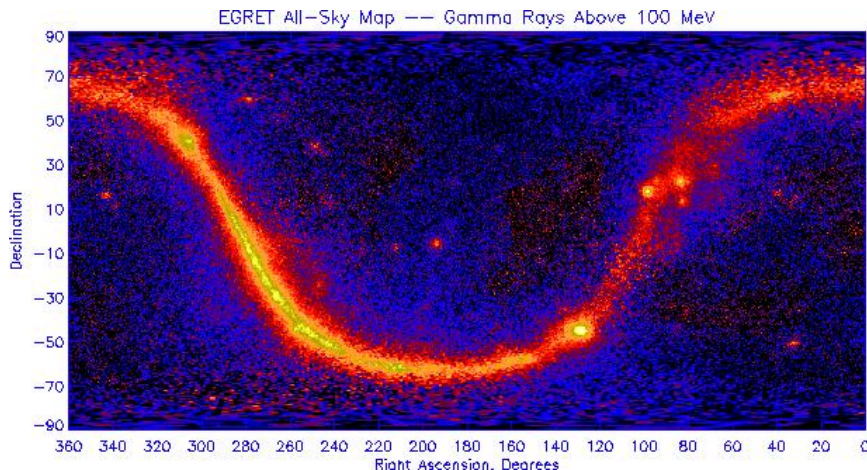


ELECTROMAGNETIC RADIATION

- **Gamma rays** -- produced by the Universe
 - Nuclear physics
 - (alpha, beta, gamma-decay, radioactive)
- **X-ray** -- produced by the Universe
 - Inner electron
 - e.g. airport security
- **Ultraviolet** -- produced by the Sun
 - Outer electron
 - Sun burn and tan



Energetic gamma-ray experiment telescope on NASA's GRO



Not used
for EO
remote
sensing

ELECTROMAGNETIC RADIATION

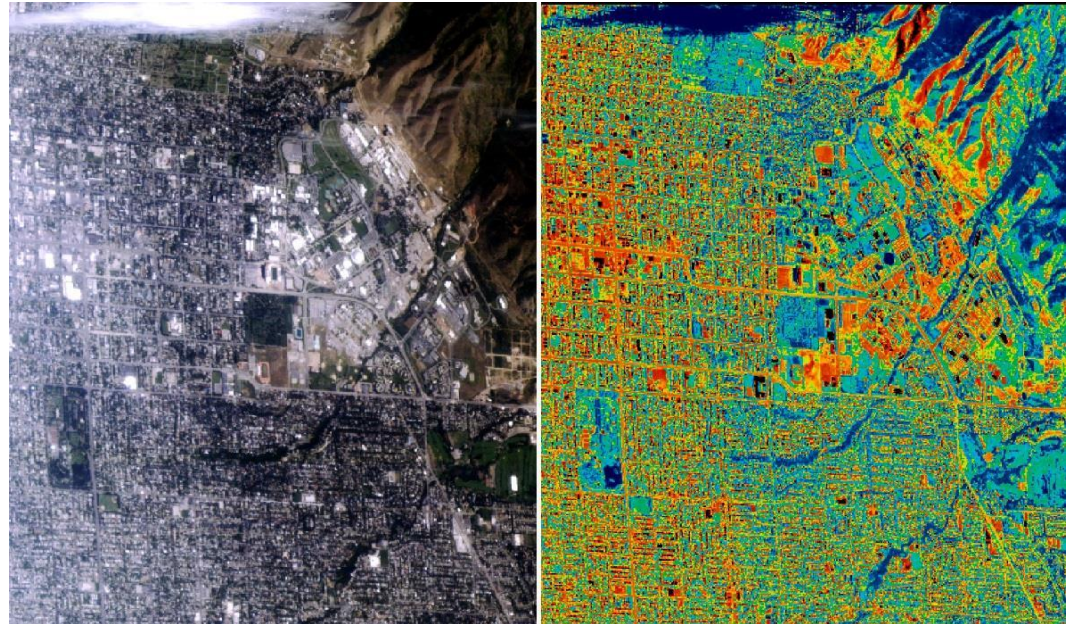


ELECTROMAGNETIC RADIATION

Used for EO remote sensing

0.6-0.7 μm 0.5-0.6 μm 0.3-0.5 μm

- **Visible** -- produced by the Sun, light bulbs, stars
 - Outer electron transition
 - Composed of blue, green and red bands
 - Vegetation, urban, snow, ocean color RS
- **Infrared** -- produced by the Earth, Sun
 - Outer electron molecular vibrations
 - LST, ET, fire scars, ocean pollution



ELECTROMAGNETIC RADIATION

- **VIS-IR**

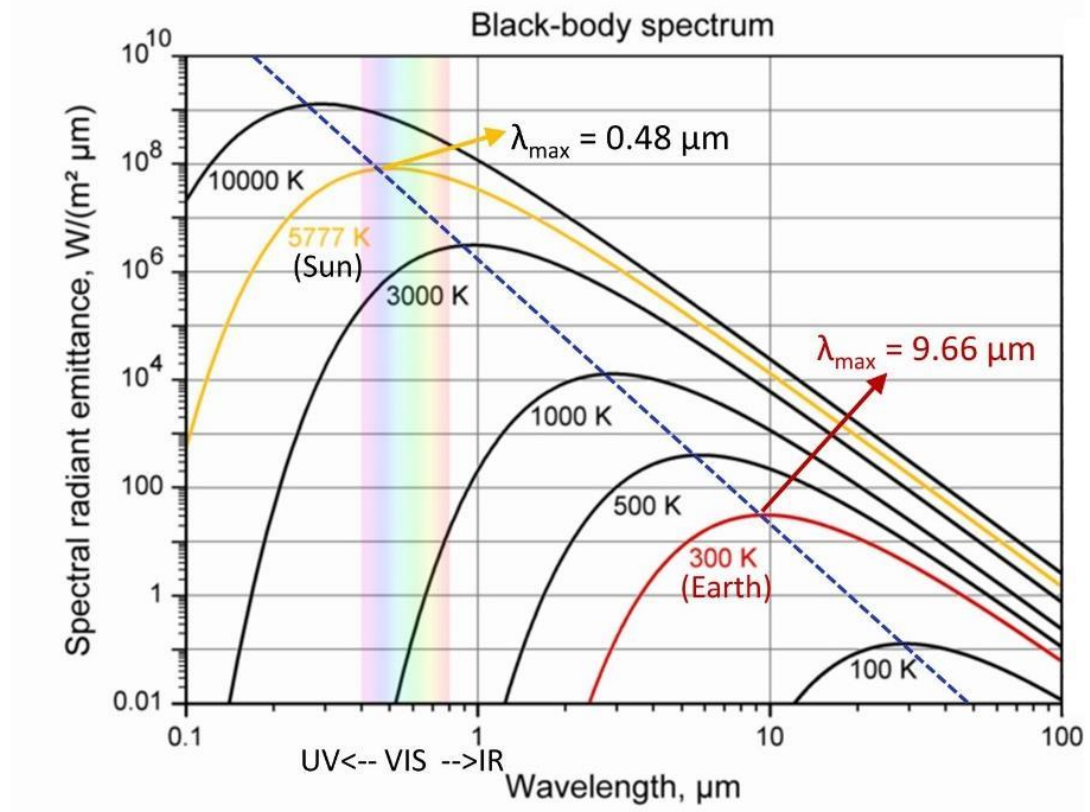
0.3-3 μm : reflective region

8-14 μm : thermal emission

3-5.5 μm : both reflection and emission

- **Thermal Earth Observation:**

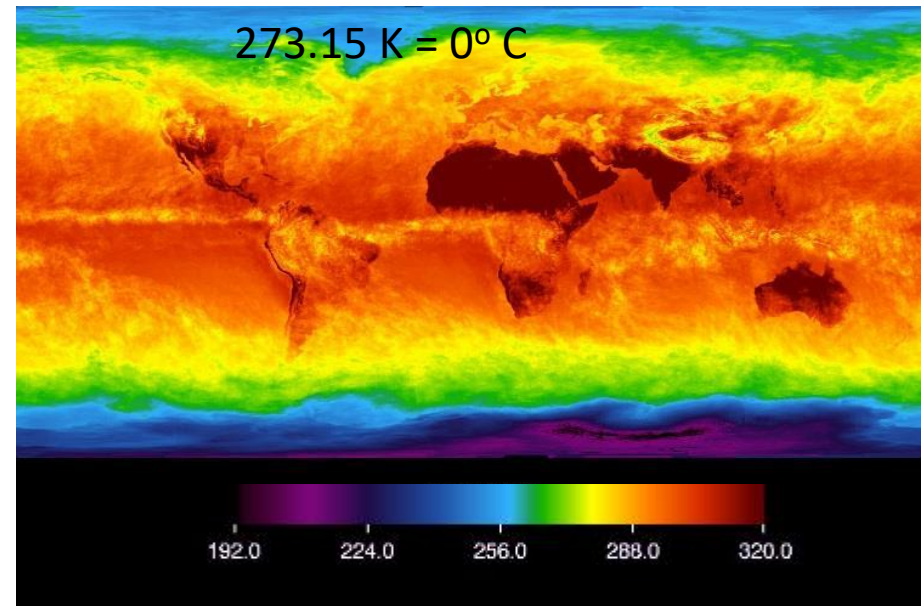
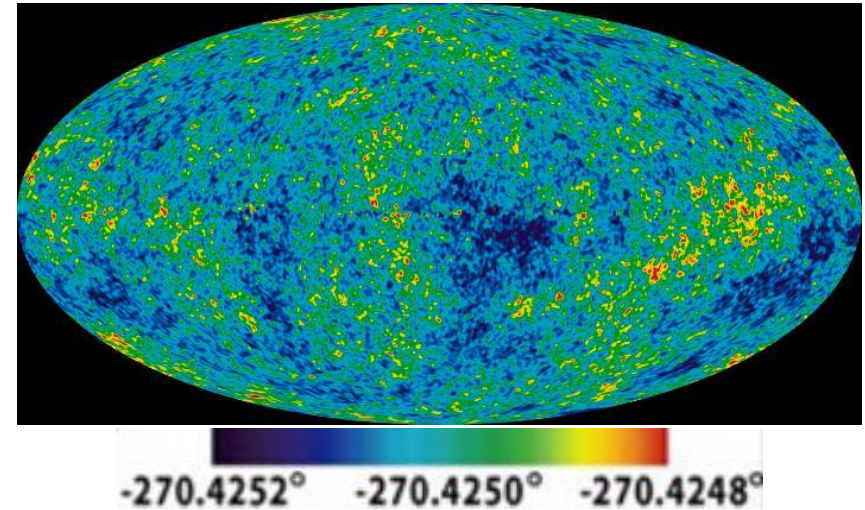
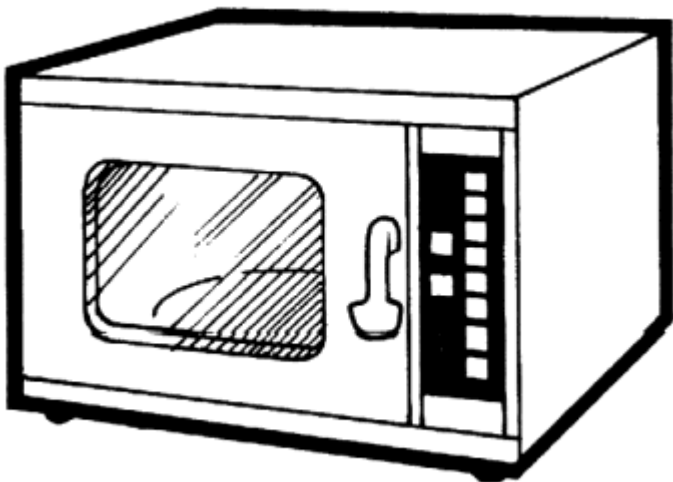
- No need for illumination
- Need to be cloud-free



ELECTROMAGNETIC RADIATION

Used for EO remote sensing

- **Microwaves** -- produced by the Earth, Cosmic background radiance (CBR)
Molecular rotations, electron spin flips
No need for illumination
No need to be cloud-free
- **Radio waves** -- produced by the stars, space
- Nuclear spin flips



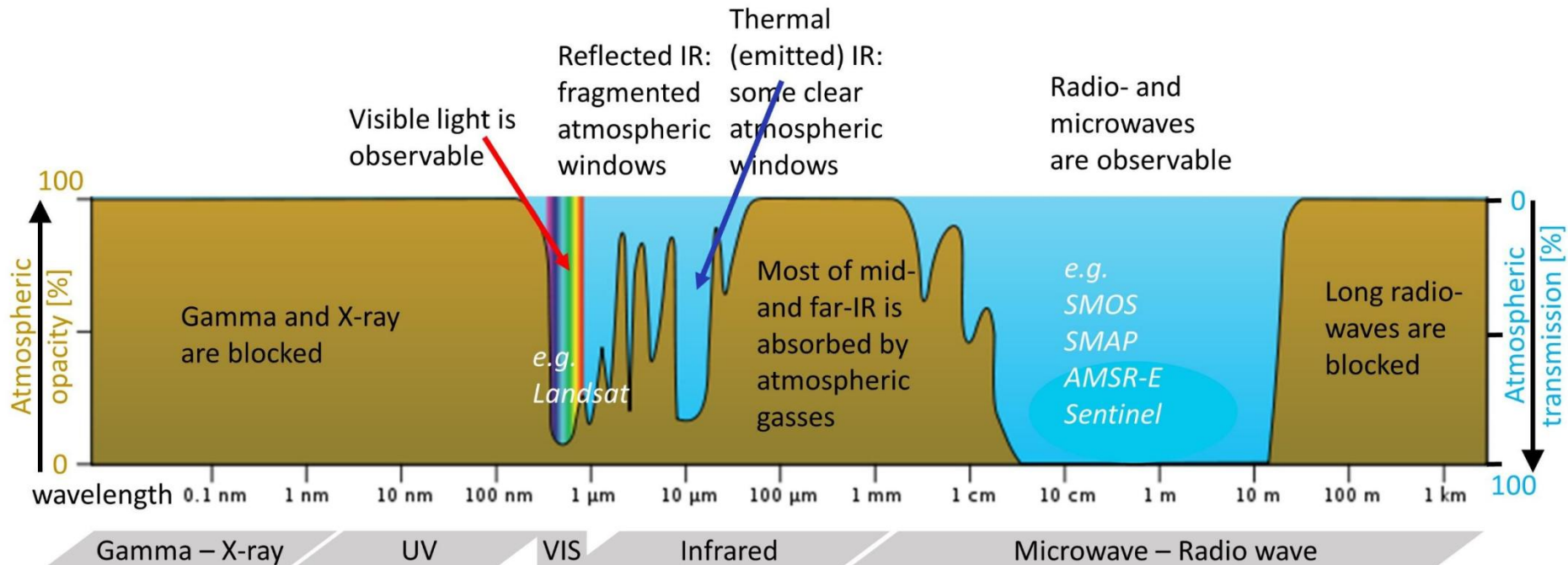
Remarks of Electromagnetic radiation

- Generated by the movement of electric charge (electron motion)
- Can be seen in particle mode or wave mode
- Quantified by intensity of photons or magnitude of wavelength
- Properties of radiation reflected or emitted off the element are used to determine objects (remote sensing)

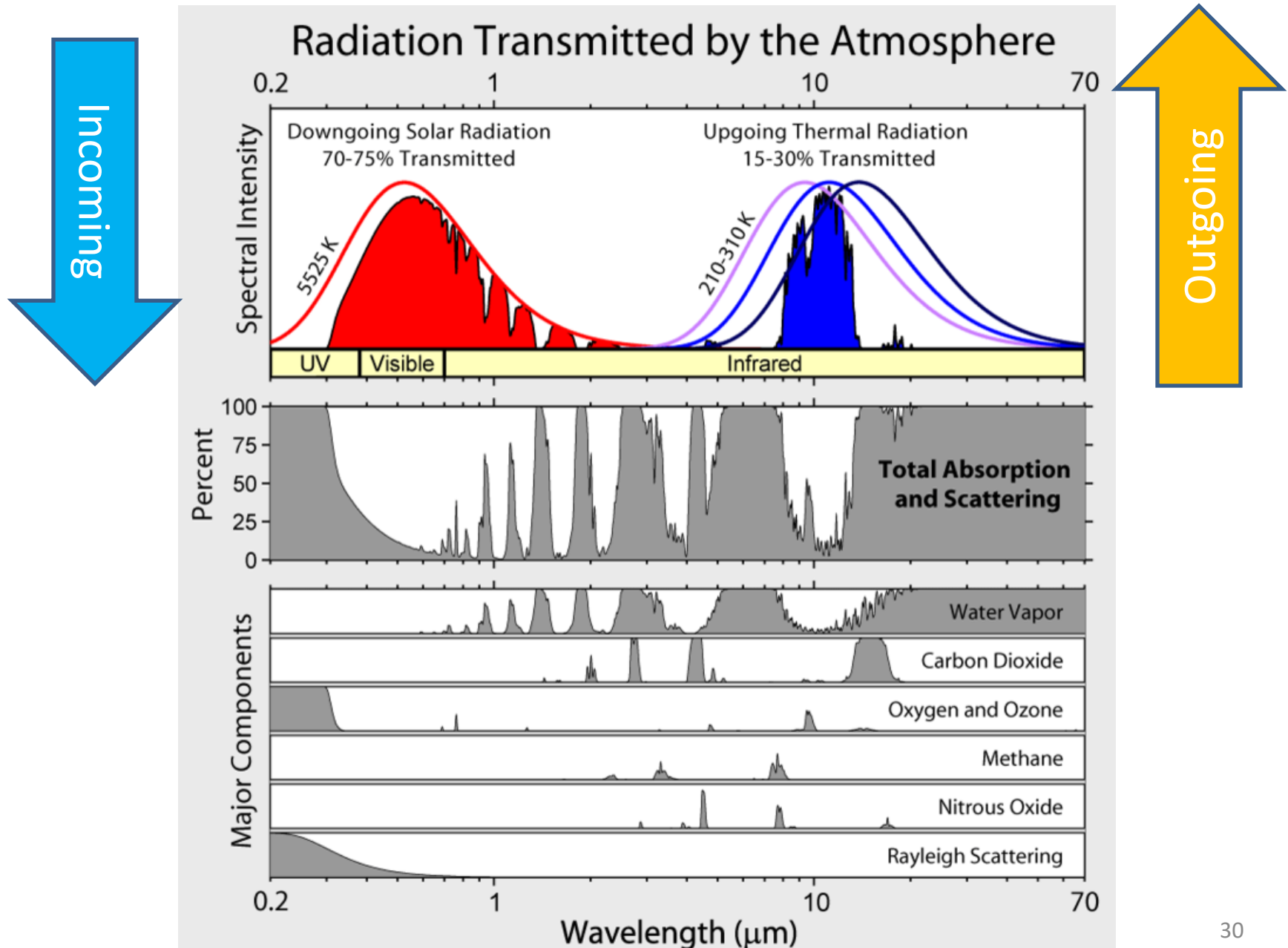
2.2 Electromagnetic radiation interacts With The Atmosphere

ATMOSPHERIC WINDOWS

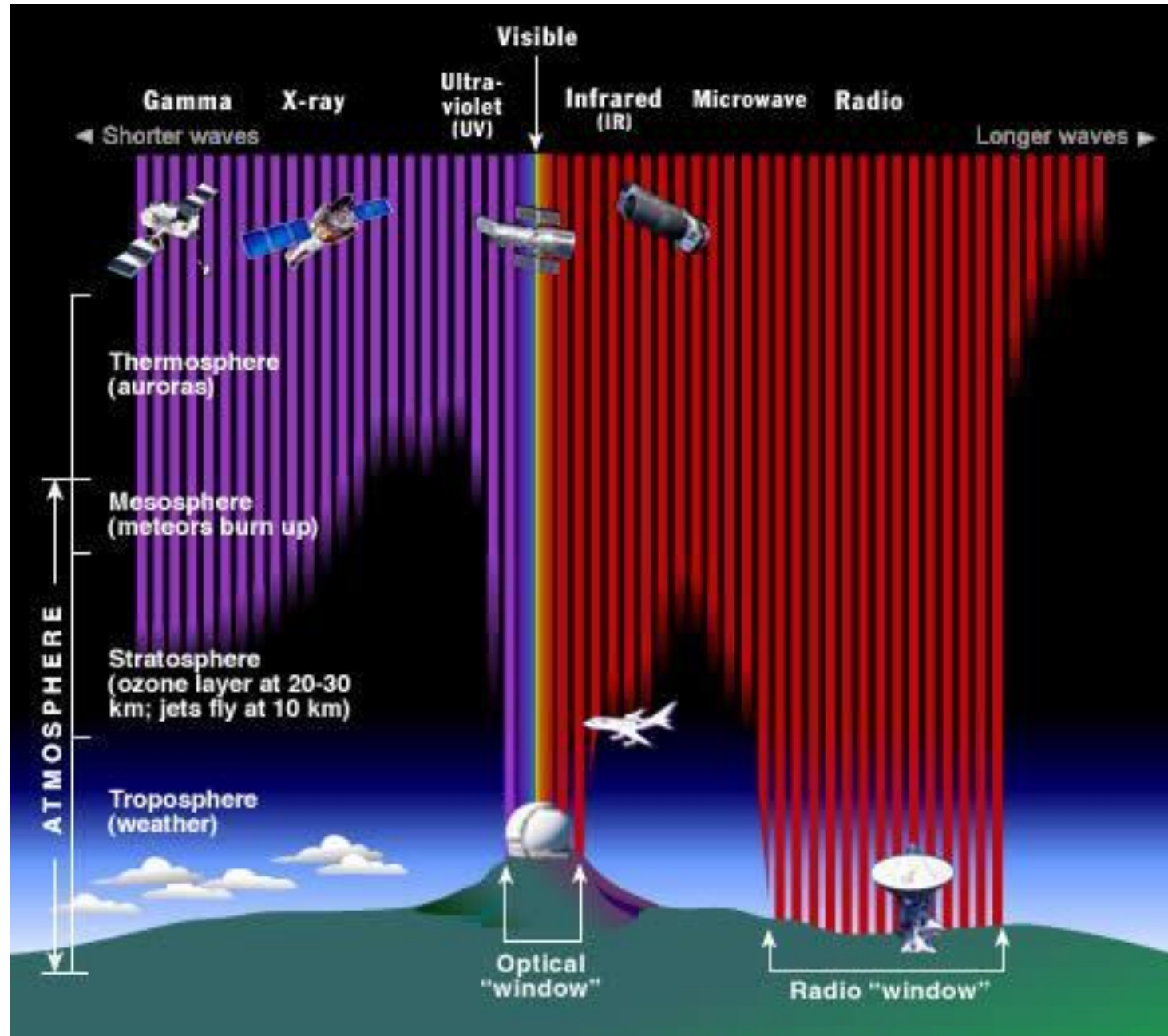
1. Optical window
2. Infrared window
3. Radio window



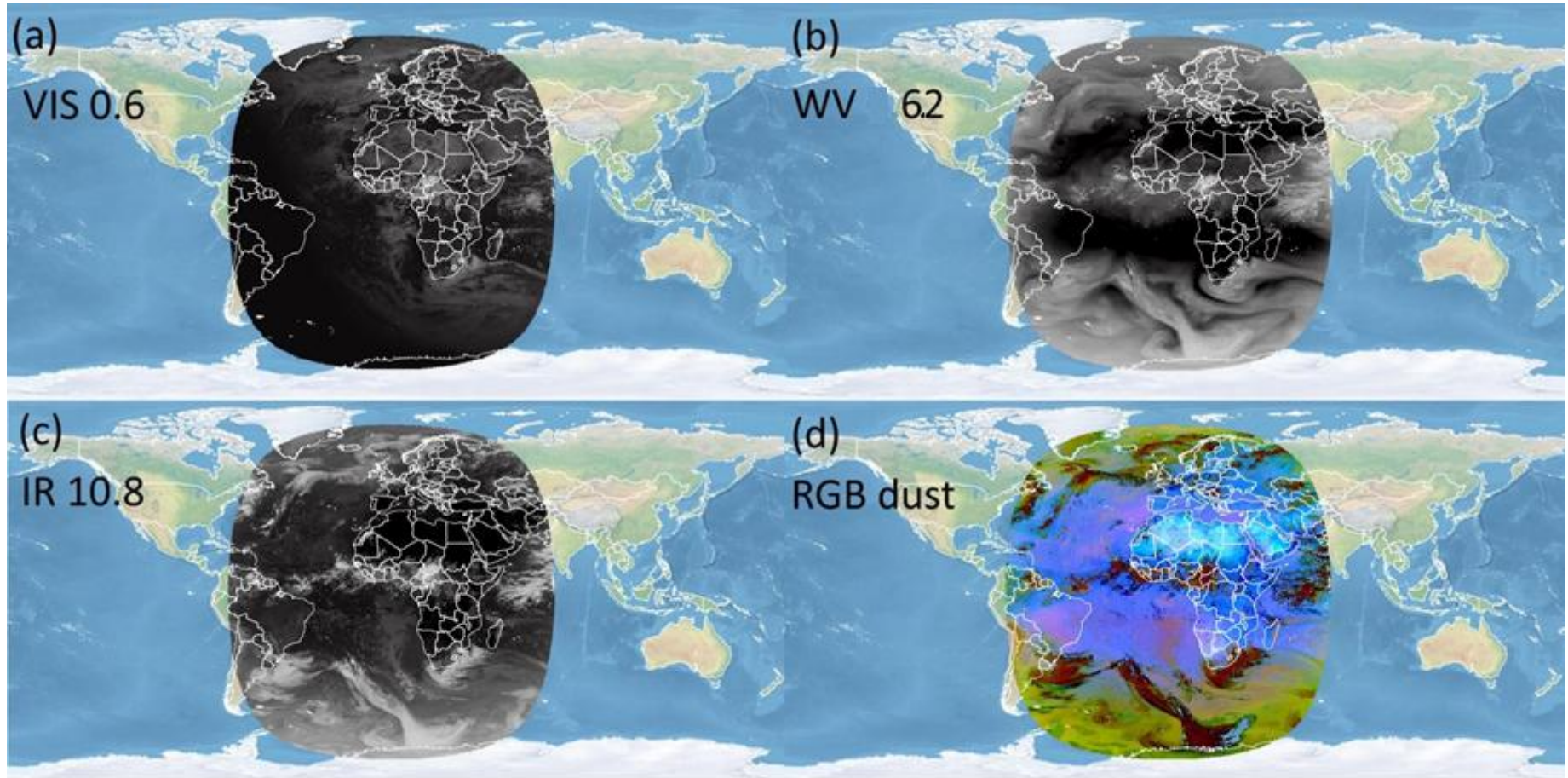
ATMOSPHERIC WINDOWS



THE OTHER WAY AROUND – LOOKING INTO SPACE



REMOTE SENSING EXAMPLE

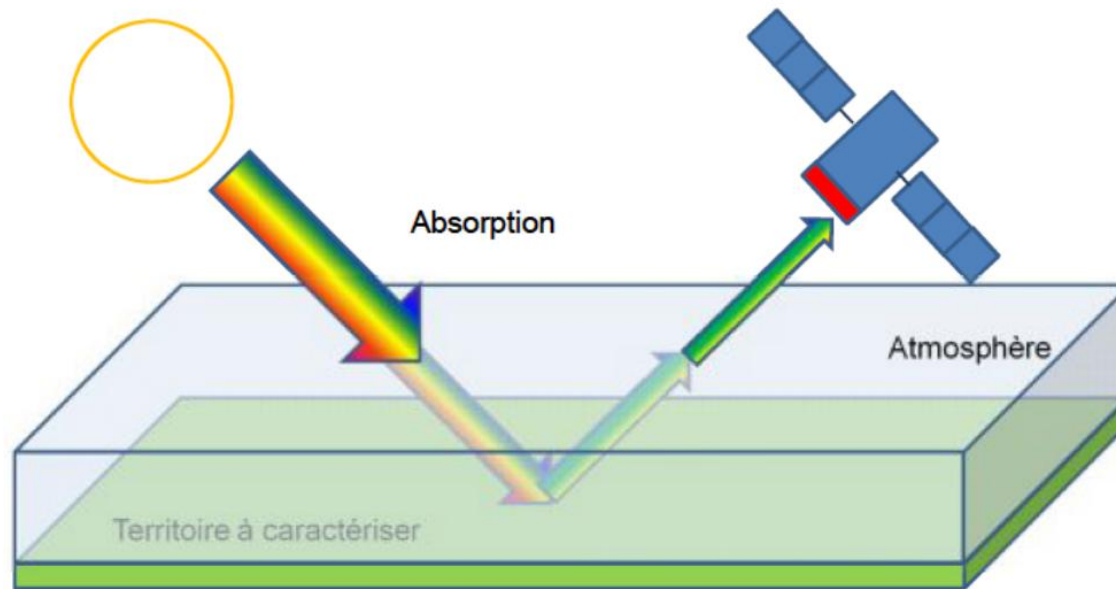


Meteosat
26 July 2016

INTERACTIONS WITH THE ATMOSPHERE

In the visible

Atmospheric contribution to take into account....

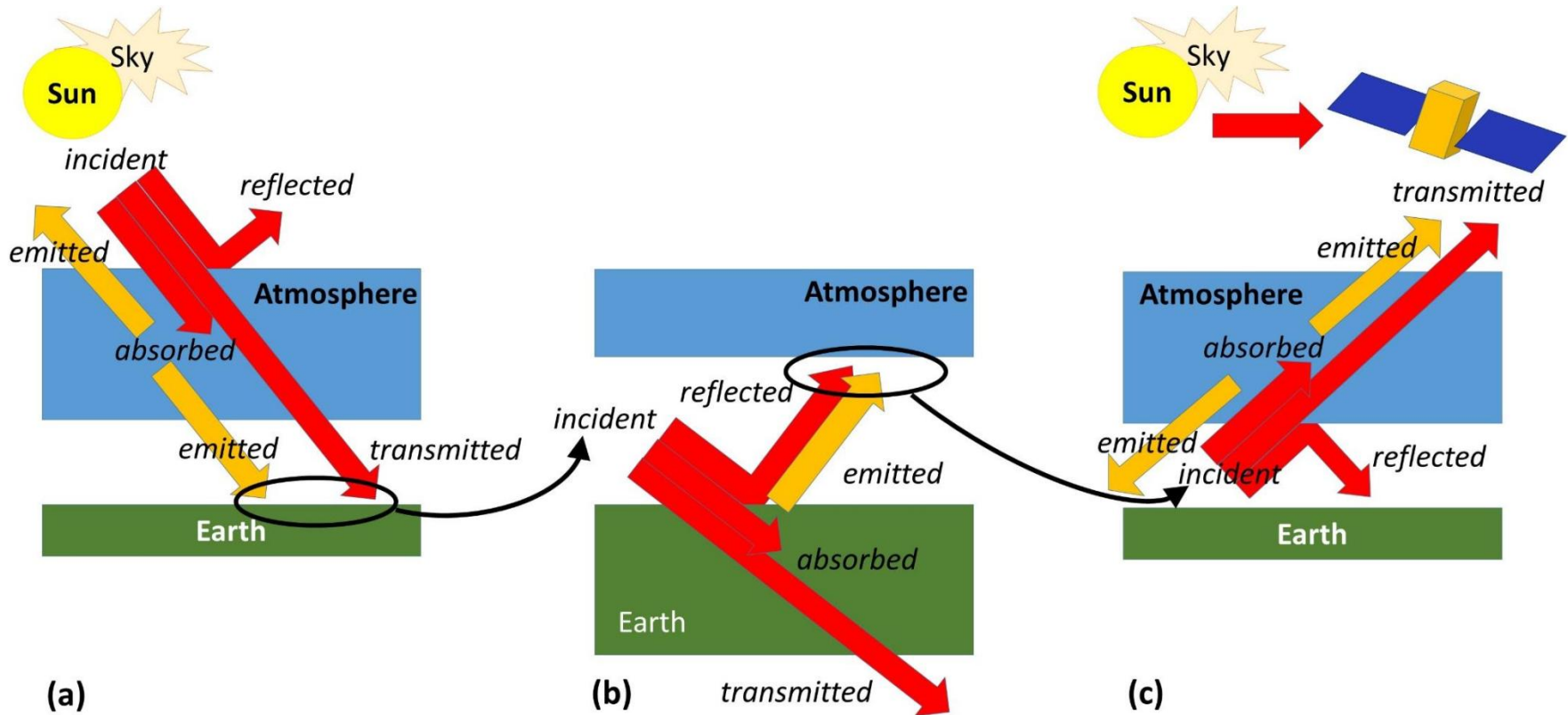


The atmosphere introduces a change in speed, direction, polarization, intensity, spectral distribution of radiation. Effect depends on atmospheric path.

INTERACTIONS WITH THE ATMOSPHERE

Radiant budget

$$\Phi_i = \Phi_{\text{reflected}} + \Phi_{\text{absorbed}} + \Phi_{\text{transmitted}}$$



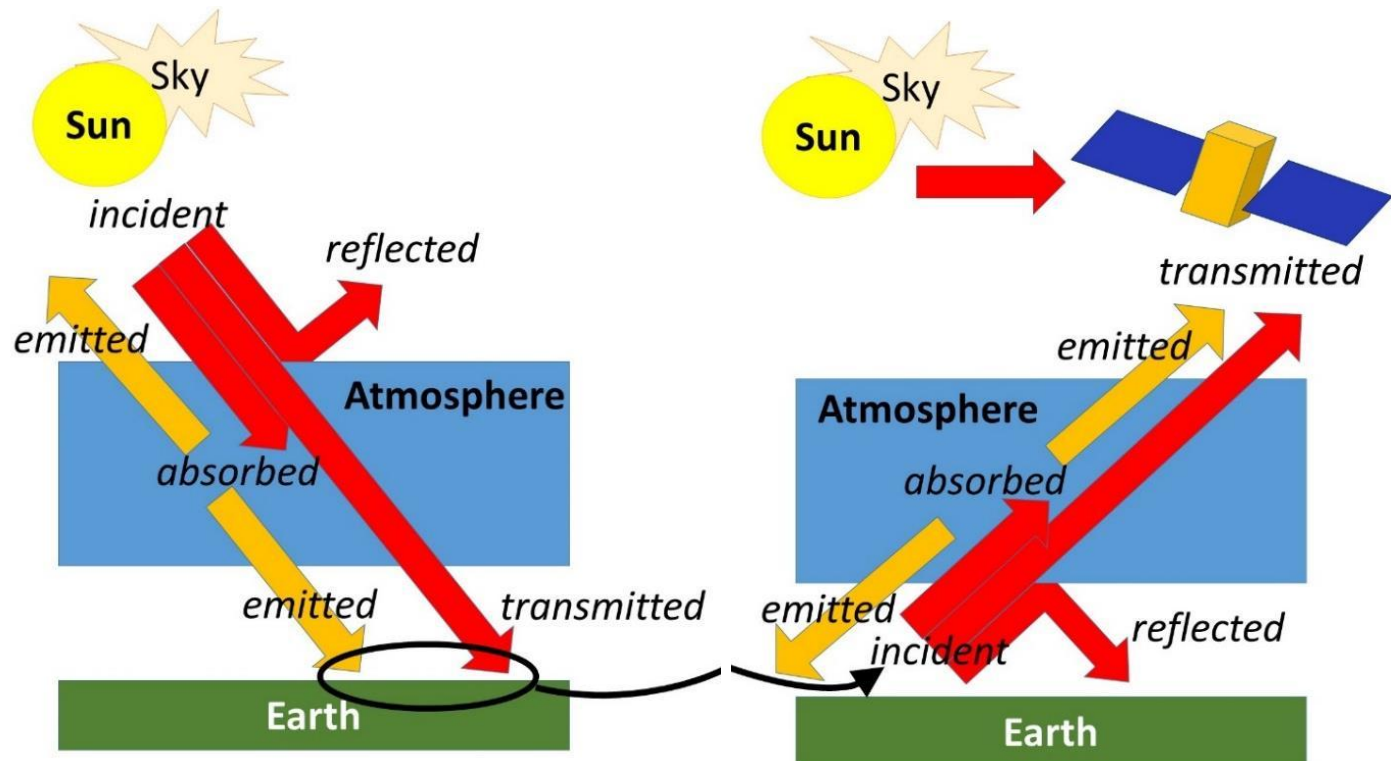
INTERACTIONS WITH THE ATMOSPHERE

Transmission, reflection:

- Refraction (*change wavelength, intensity, but frequency of radiation unchanged*)
- Scattering (*Wavelength, Frequency of radiation unchanged*)

Absorption, emission (*Change in frequency*)

Atmospheric properties



TRANSMISSION

- Pass through medium without significant attenuation

Transmittance:

$$\tau_{\lambda} = \frac{\Phi_T}{\Phi_I}(\lambda) \quad \leq 1$$

- **Optical depth:**

$$\tau_{\lambda}^* = \ln\left(\frac{\Phi_I}{\Phi_T}(\lambda)\right) = -\ln(\tau_{\lambda}) \quad \geq 0$$

- **Absorbance:** $A = \log_{10}\left(\frac{\Phi_I}{\Phi_T}\right) [-]$

varies with composition & depth of medium, temperature, humidity, wavelength

$$\tau_{\lambda} = e^{-\tau_{\lambda}^*} = 10^{-A}$$

ABSORPTION

Effective loss of energy; transformation into another type of energy

Absorptance:

$$\alpha_{\lambda} = \frac{\Phi_A}{\Phi_I}(\lambda)$$

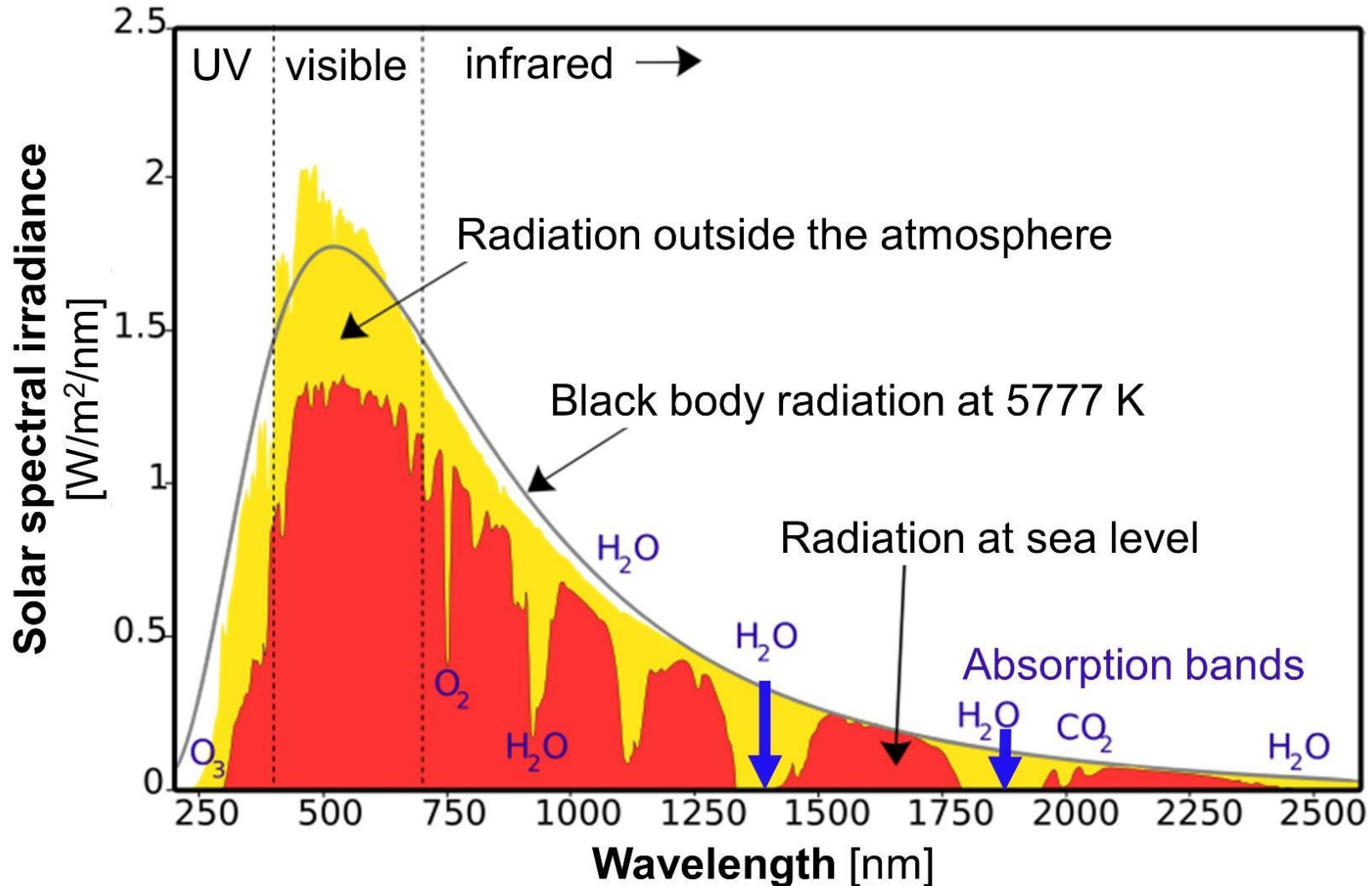
varies with composition & depth of medium, temperature, humidity, wavelength

**Atmospheric
windows**

ABSORPTION

Sun is our source of light

Atmosphere gets in the way



REFLECTION

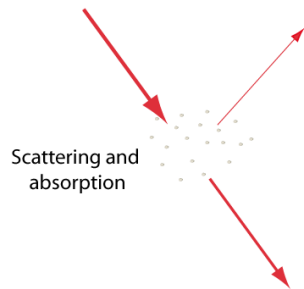
Either at **surface** or within medium

Reflectance:

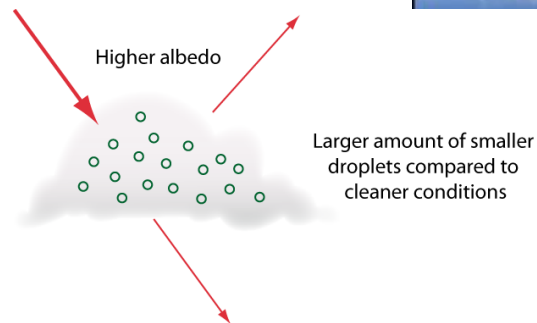
$$\rho_{\lambda} = \frac{\Phi_R}{\Phi_I}(\lambda)$$

Albedo:

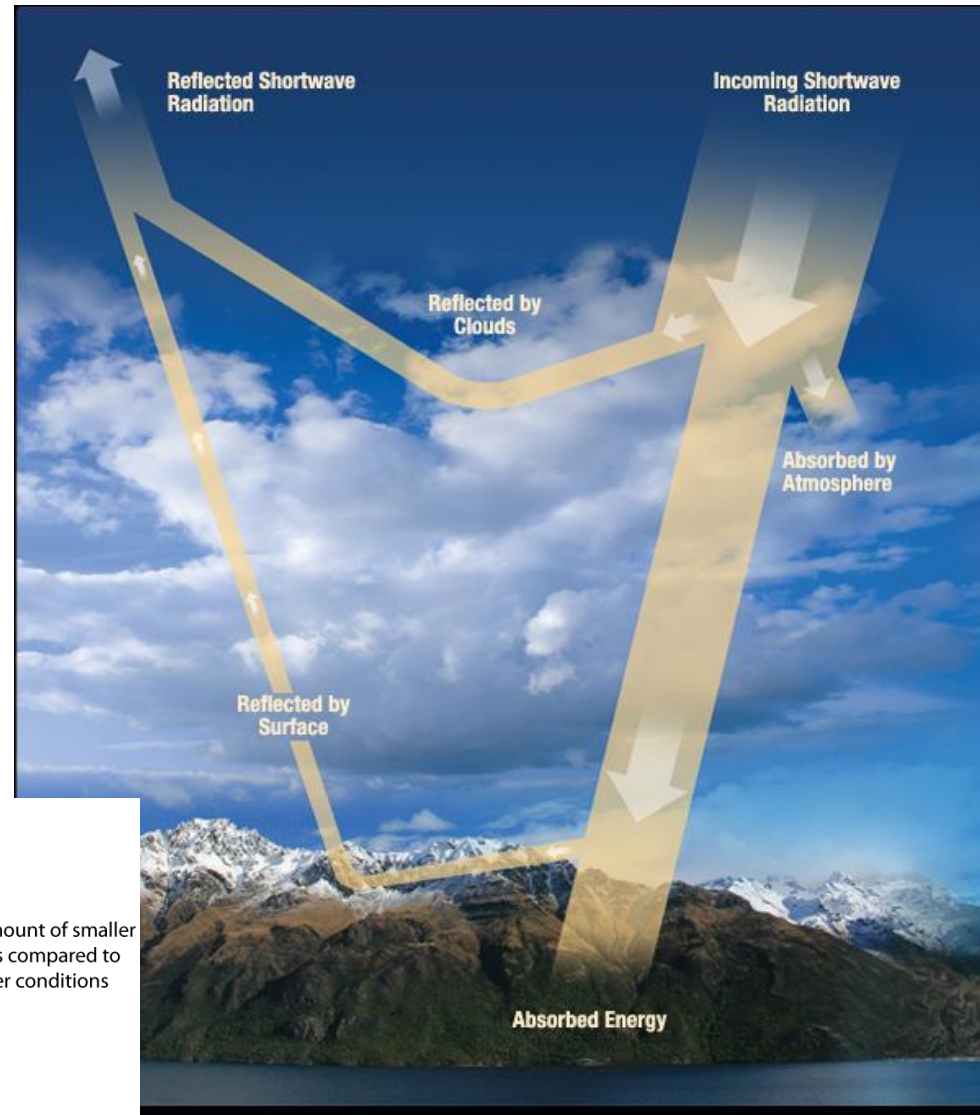
directional integration
cloud albedo



Direct effects

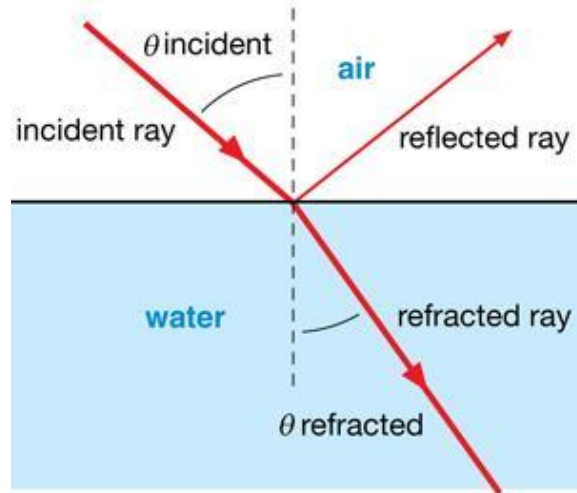


First indirect effect: cloud-albedo effect



REFRACTION

- Transmission through a **surface**: change in density



- **Snell's law:**

$$\frac{n_1}{n_2} = \frac{\theta_T}{\theta_I}$$

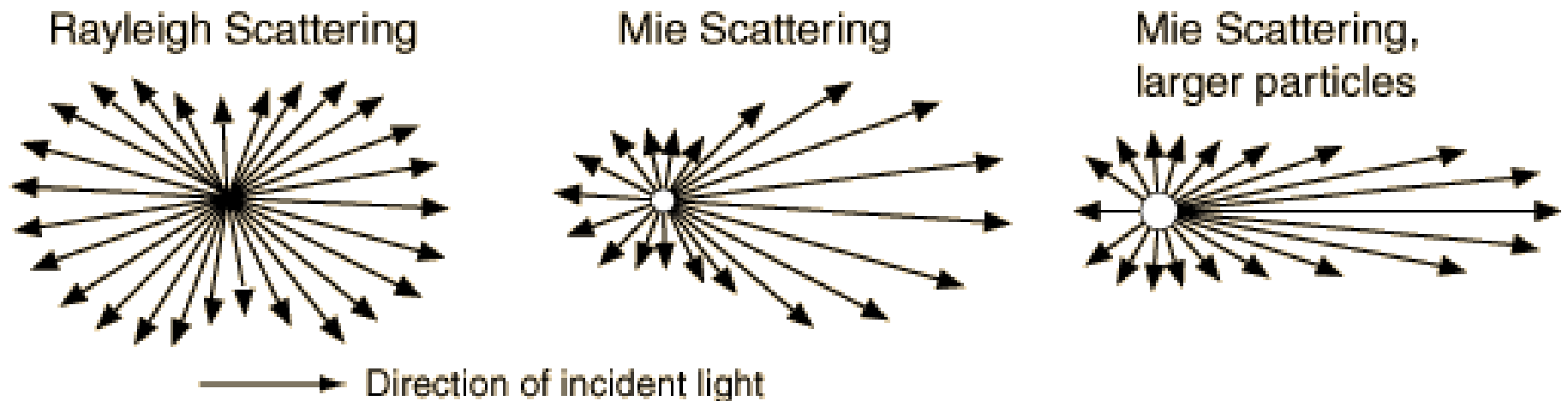
- **Fresnel law:**

- Refracted energy fraction is given by transmittance (unrelated to transmission *through* a medium)
- Phase shift

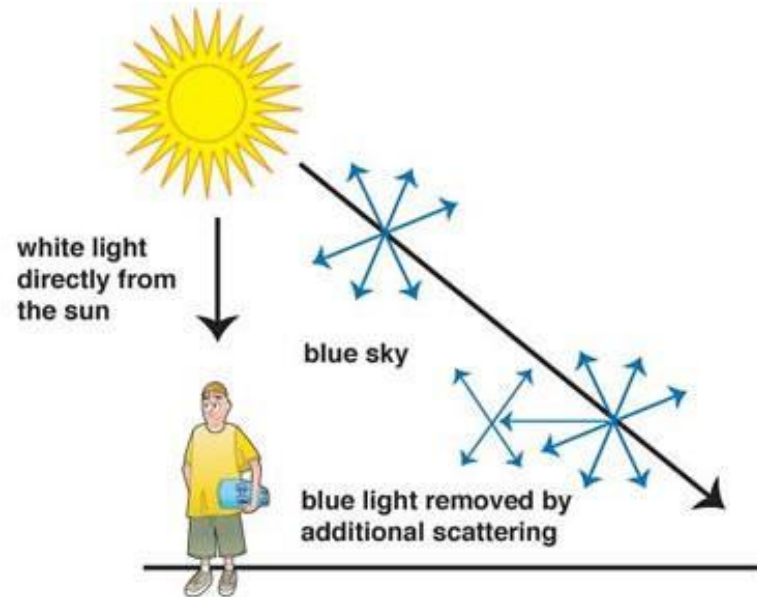
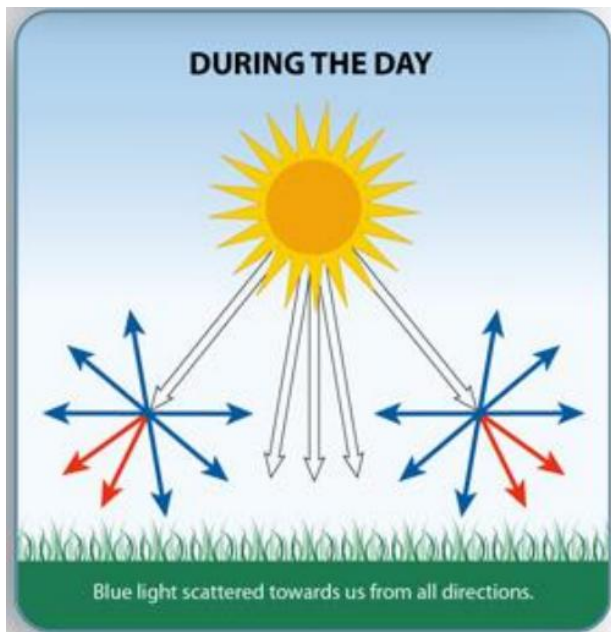
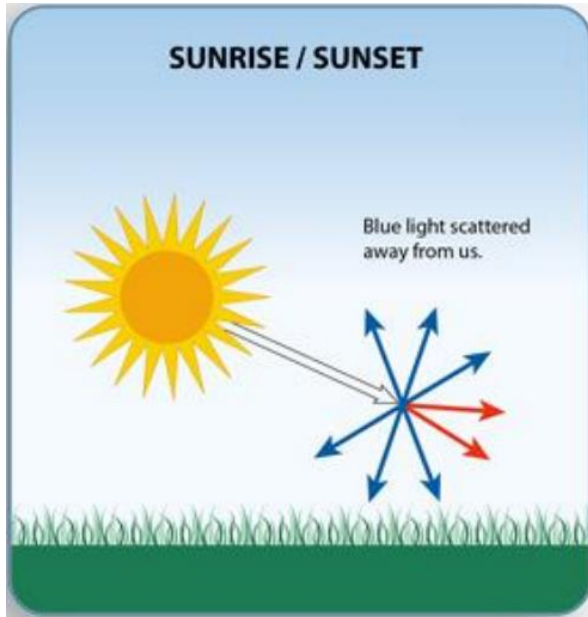
SCATTERING

Depending on the wavelength of the radiation, and the relative size of the scattering particles:

- Rayleigh scattering (particles smaller than EMR wavelengths)
 - Air molecules, N_2 , O_2
- Mie scattering (particles similar size as EMR wavelengths)
 - Dust
- Non-selective scattering
 - Water in clouds



SCATTERING



ATTENUATION

Absorption
Reflection
Scattering

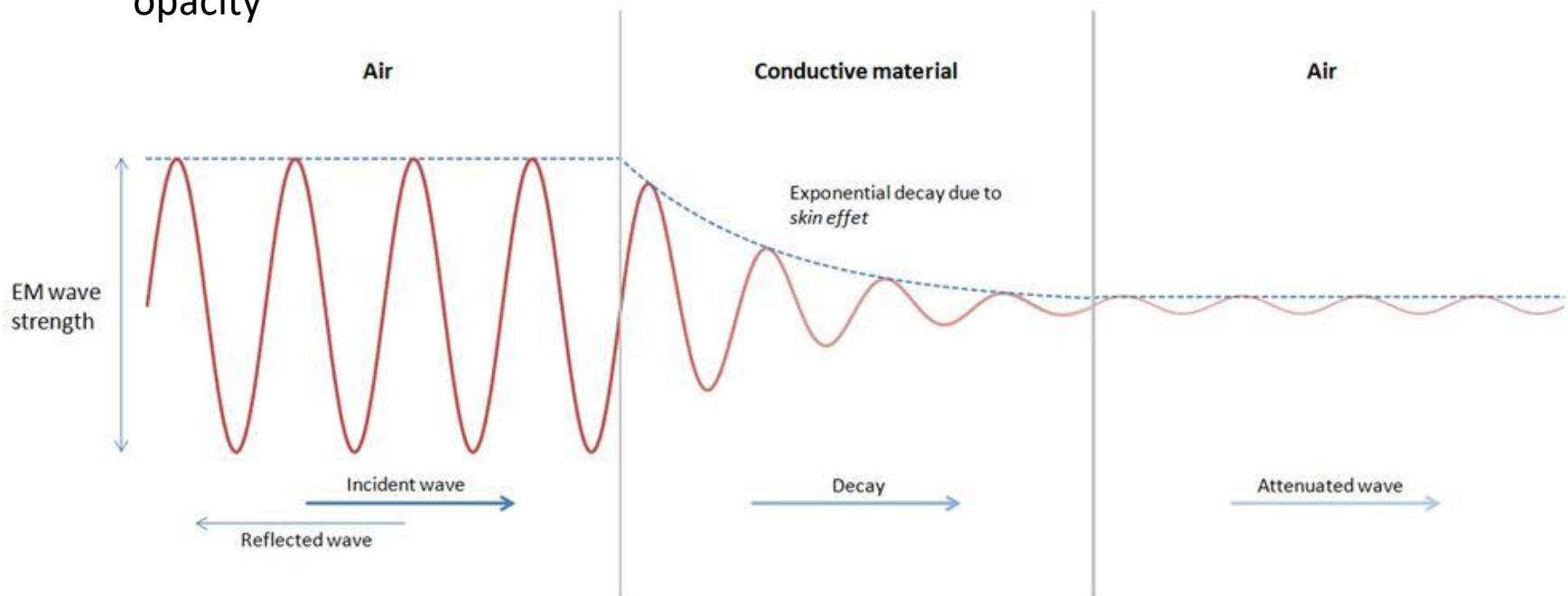
Radiant flux reduction due to
passage through a medium

*Opacity or
absorption coefficient*

*Path through
atmosphere*

$$\Phi_T(\lambda) = \Phi_I(\lambda) \cdot \exp(-k_\lambda \rho \cdot x) = \Phi_I(\lambda) \cdot \exp(-\tau_\lambda^*) = \Phi_I(\lambda) \cdot \tau_\lambda$$

extinction coefficient
opacity



EMISSION

Kirchhoff's law:

For an arbitrary body emitting and absorbing thermal radiation in thermodynamic equilibrium, the emissivity is equal to the absorptivity.

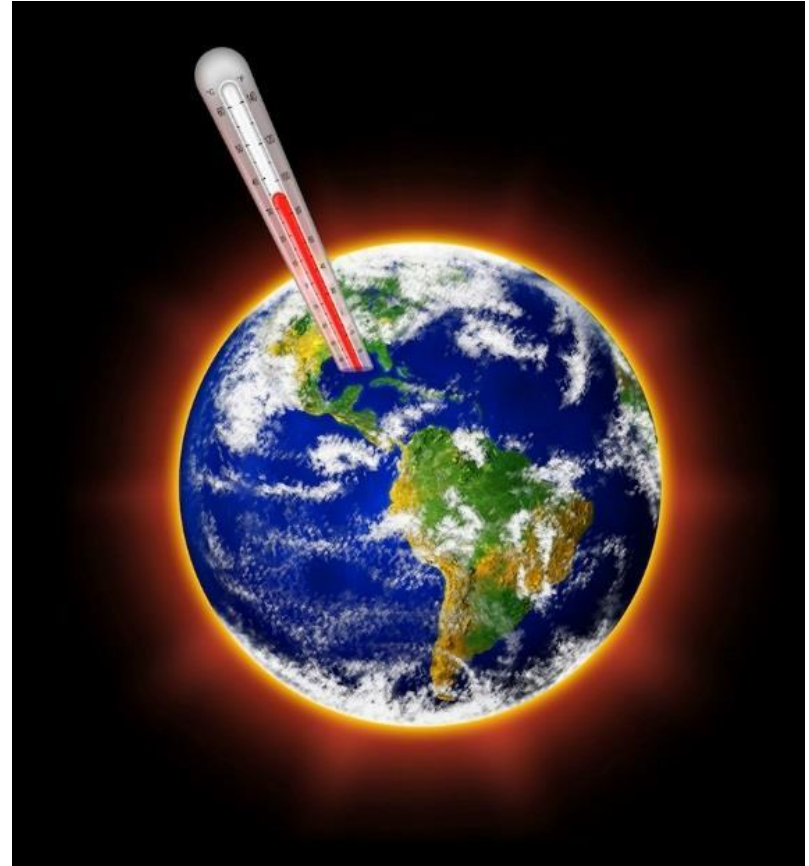
$$\epsilon_{\lambda} \sim \alpha_{\lambda} = \frac{\Phi_A}{\Phi_I}(\lambda)$$

Incoming solar radiation is absorbed and re-emitted by greenhouse gases.

$$M_{\lambda} = \epsilon_{\lambda} \cdot \sigma \cdot T^4$$

Blackbody → gray body:

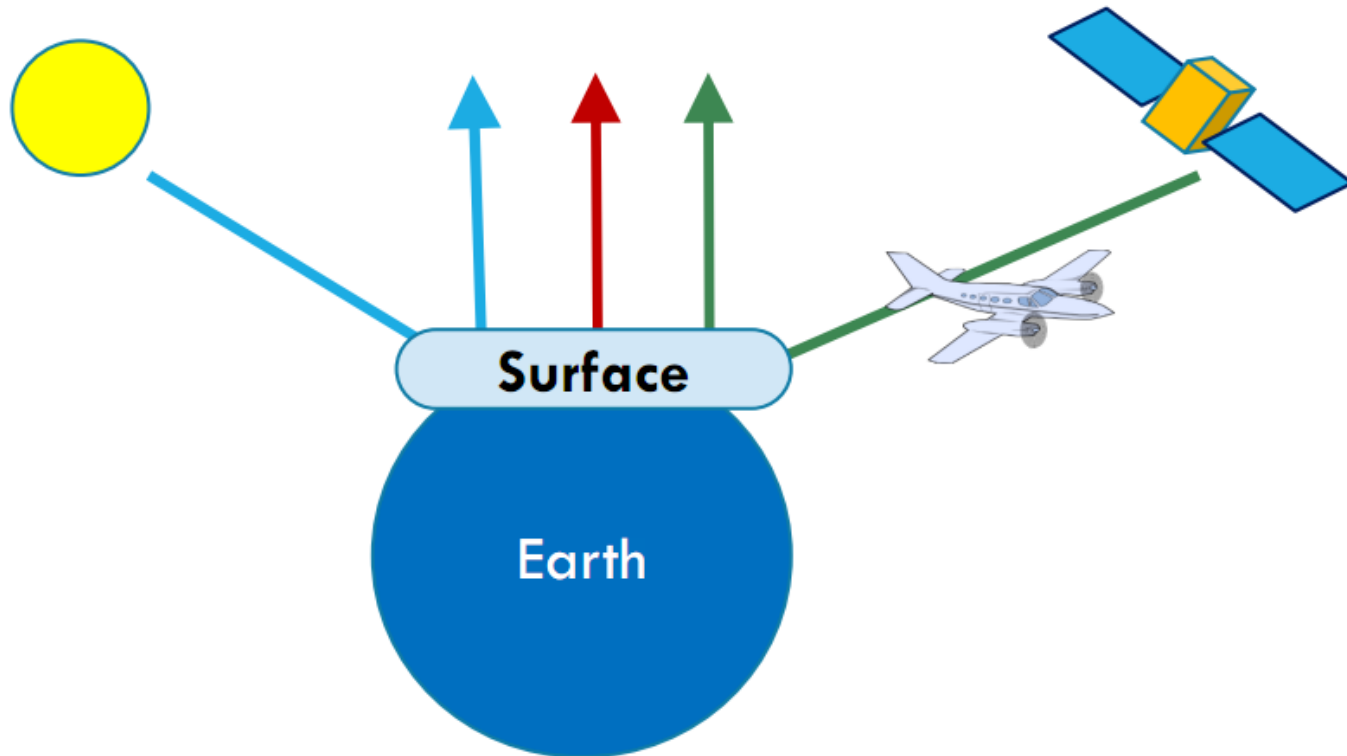
$(\alpha_{\lambda} < 1)$ are denoted as gray body emitters $(\epsilon_{\lambda} < 1)$



Remarks

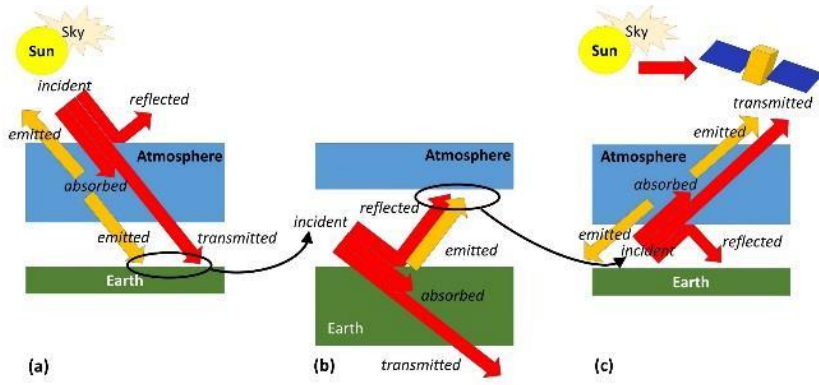
- Atmospheric windows are resulted by interaction between Electromagnetic waves with particles in atmosphere
- Interaction => Scattering: Rayleigh scattering, Mie scattering, Non-selective scattering
- Atmospheric windows allow specific Electromagnetic waves going through
- Effect of atmospheric particles may seriously impacts to the land surface investigation using remote sensing
- Some satellite system developed to investigate atmosphere.

2.3 INTERACTIONS AT THE EARTH'S SURFACE



The surface introduces a change in speed, direction, polarization, intensity, spectral distribution of radiation

INTERACTIONS AT THE EARTH'S SURFACE

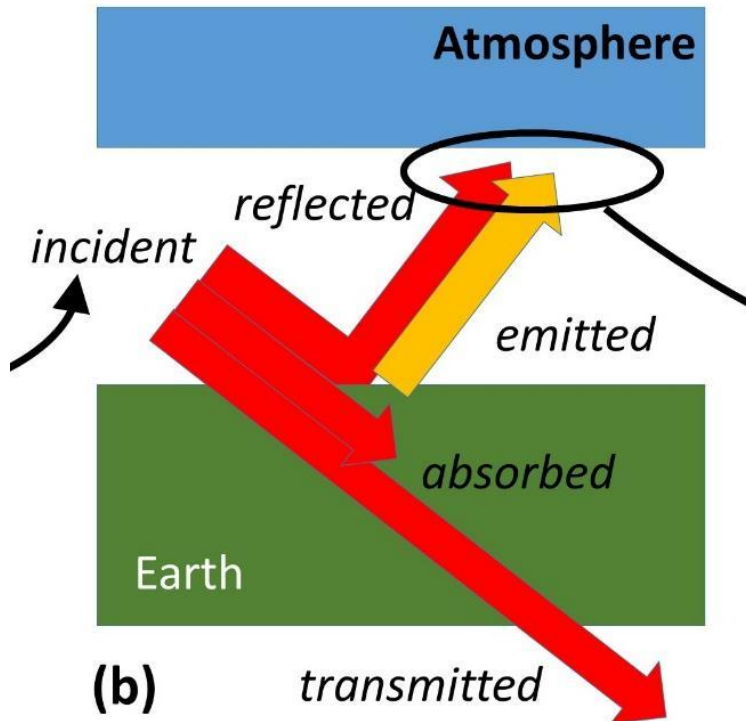


**Transmission, reflection
- Scattering**

Absorption emission

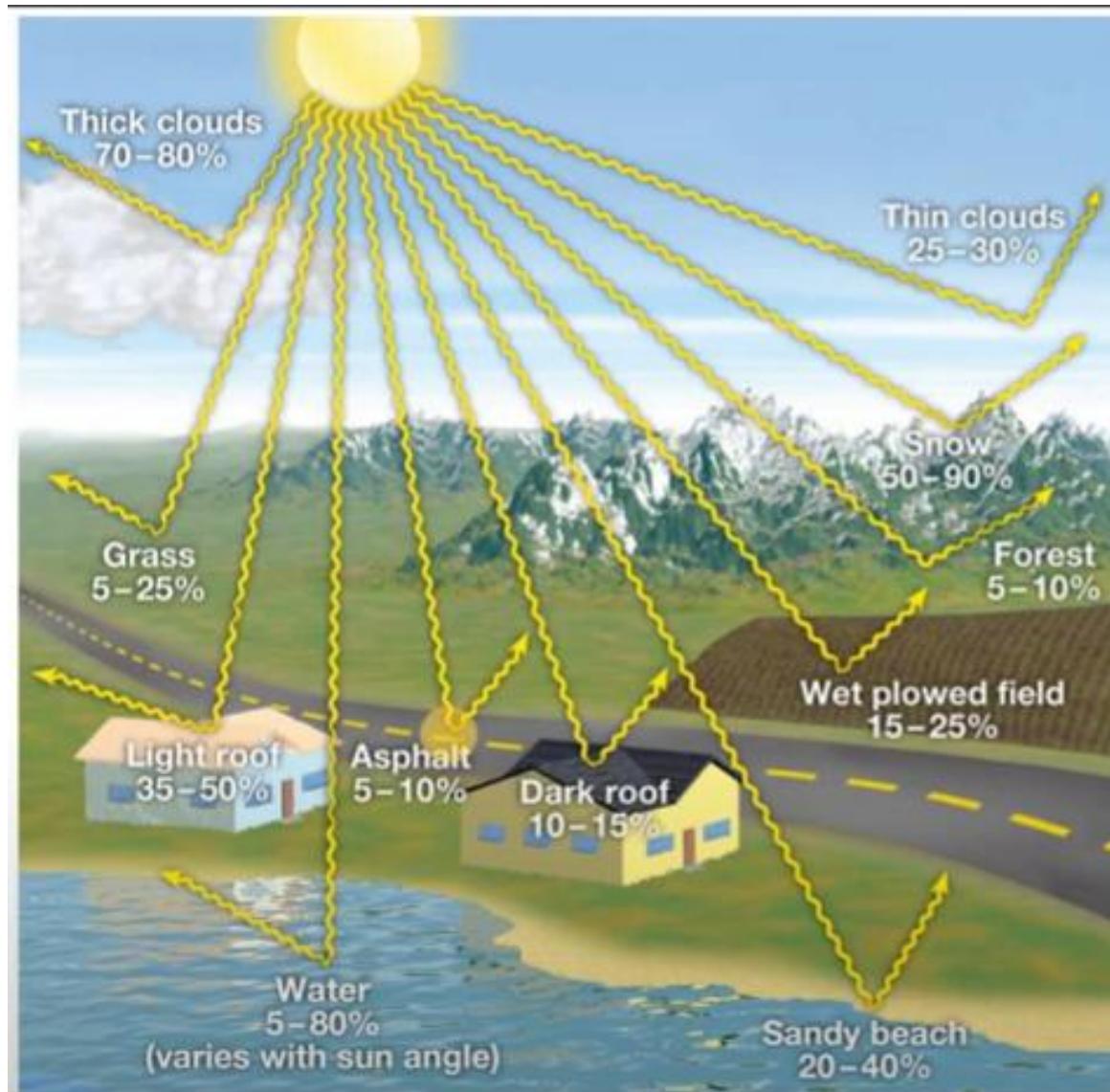
*Frequency of
radiation unchanged*

*Change in frequency,
direction, magnitude,
polarization...*



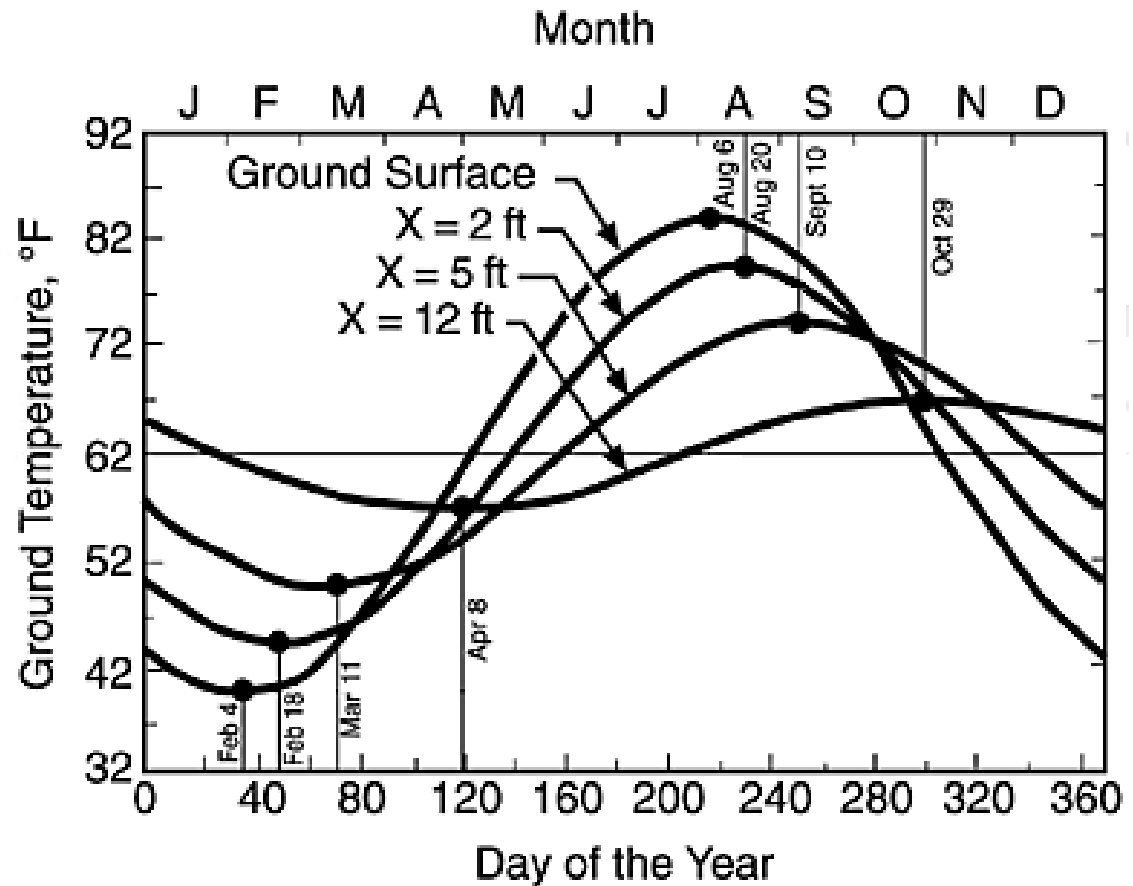
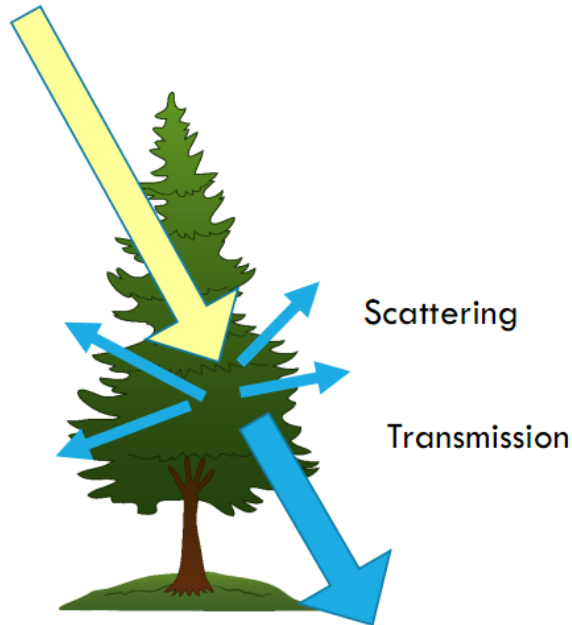
Surface properties

INTERACTIONS AT THE EARTH'S SURFACE



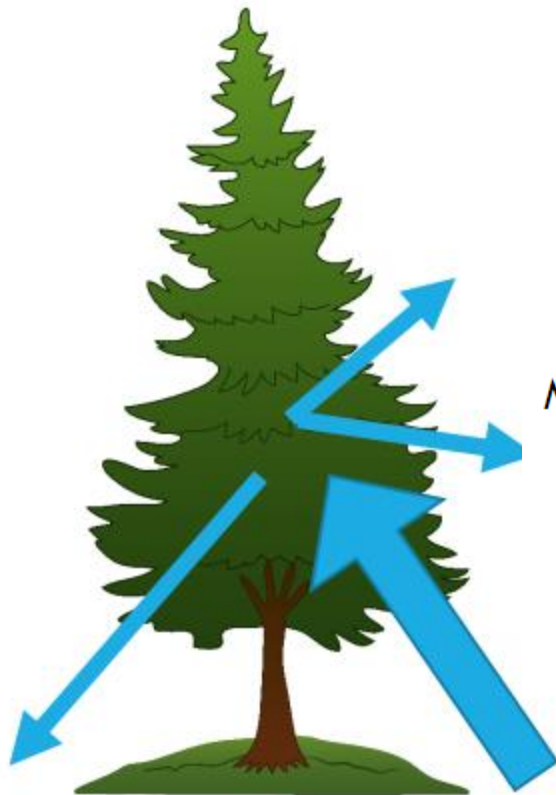
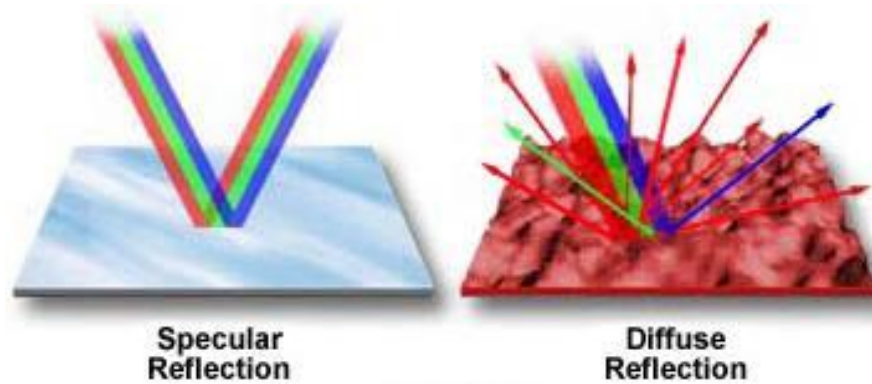
TRANSMISSION, ABSORPTION

Absorption >> warm up of the soil >> emission



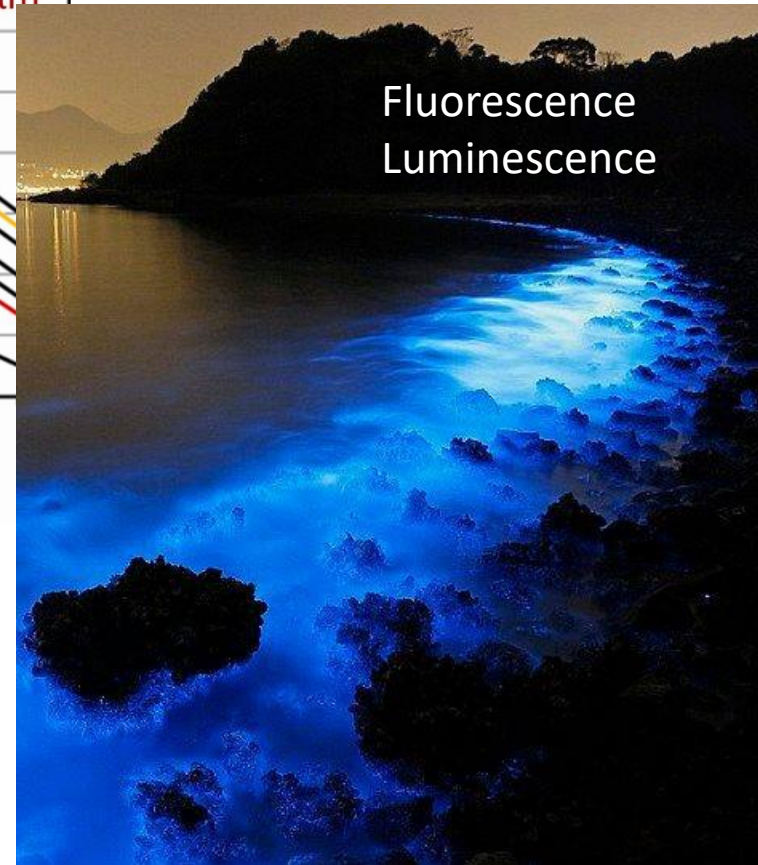
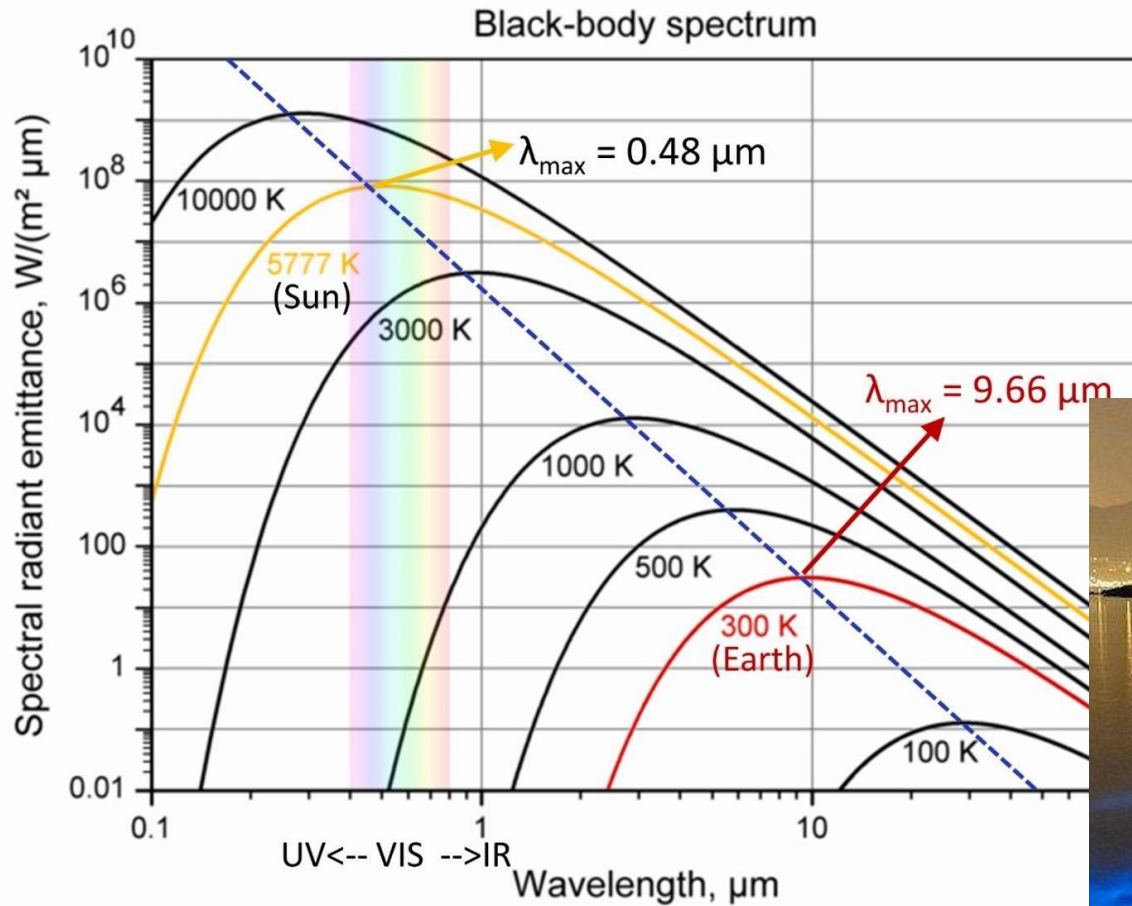
Temperature variation at different depth of the earth 49

REFLECTION

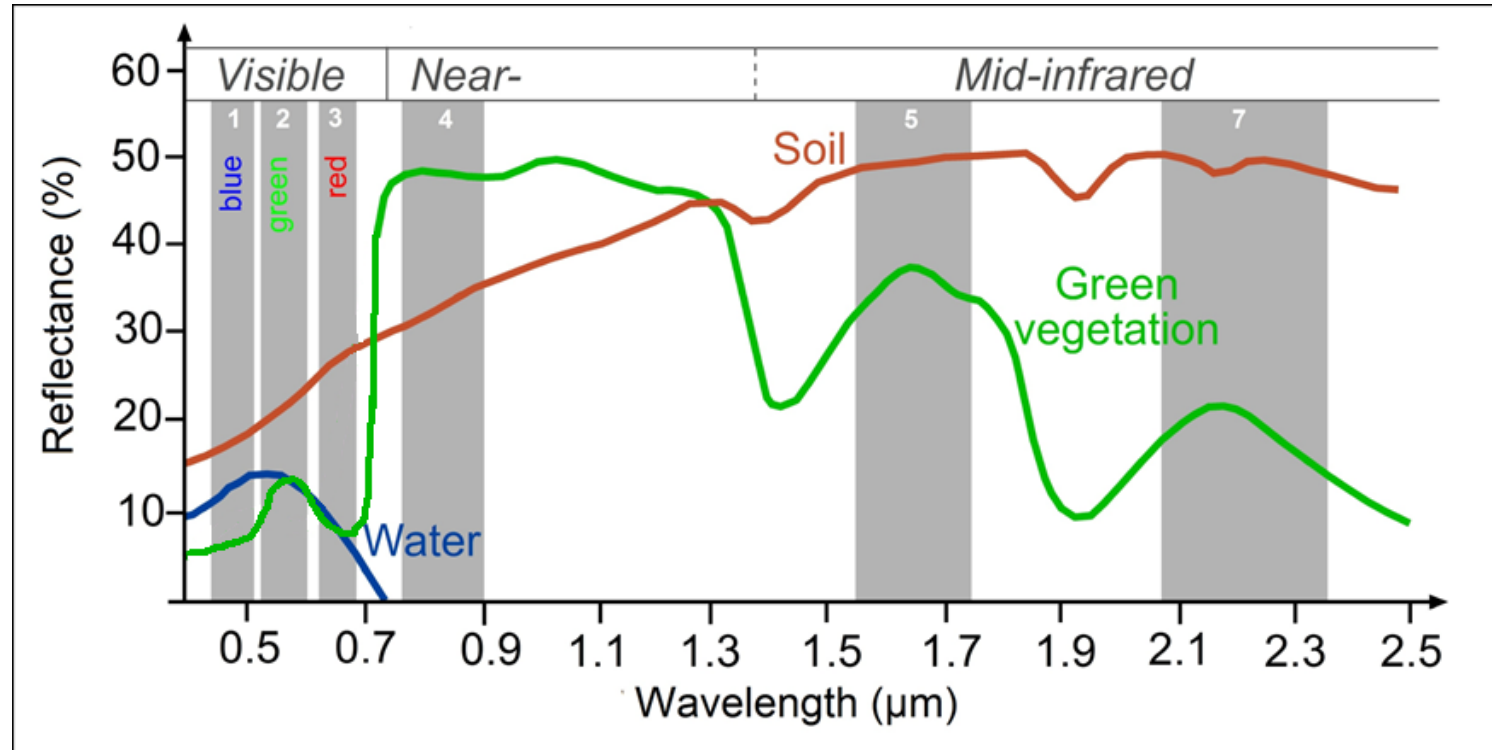


- **Specular:** smooth surface
- **Diffuse (Lambertian):** rough surface

EMISSION

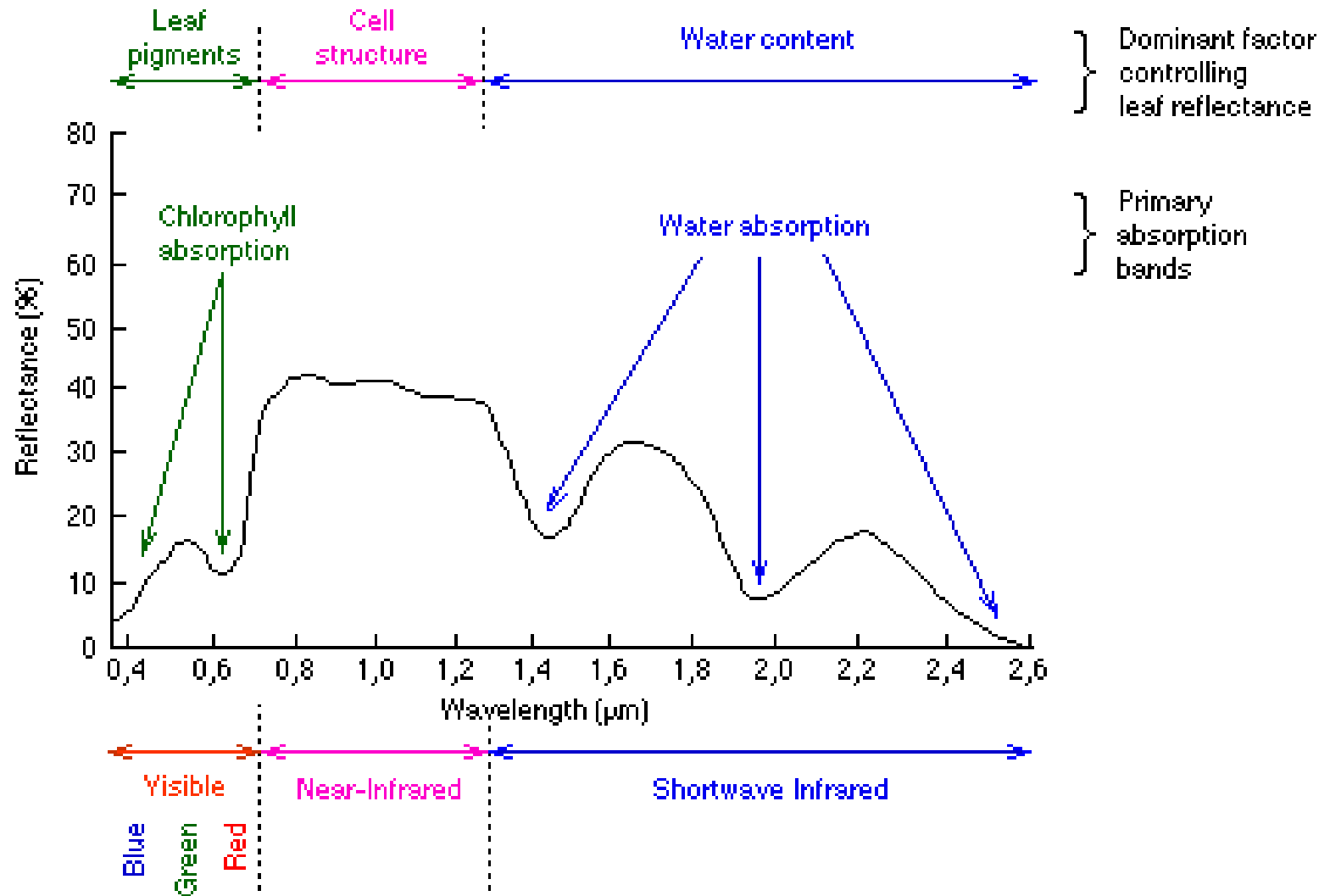


Surface spectral reflectance

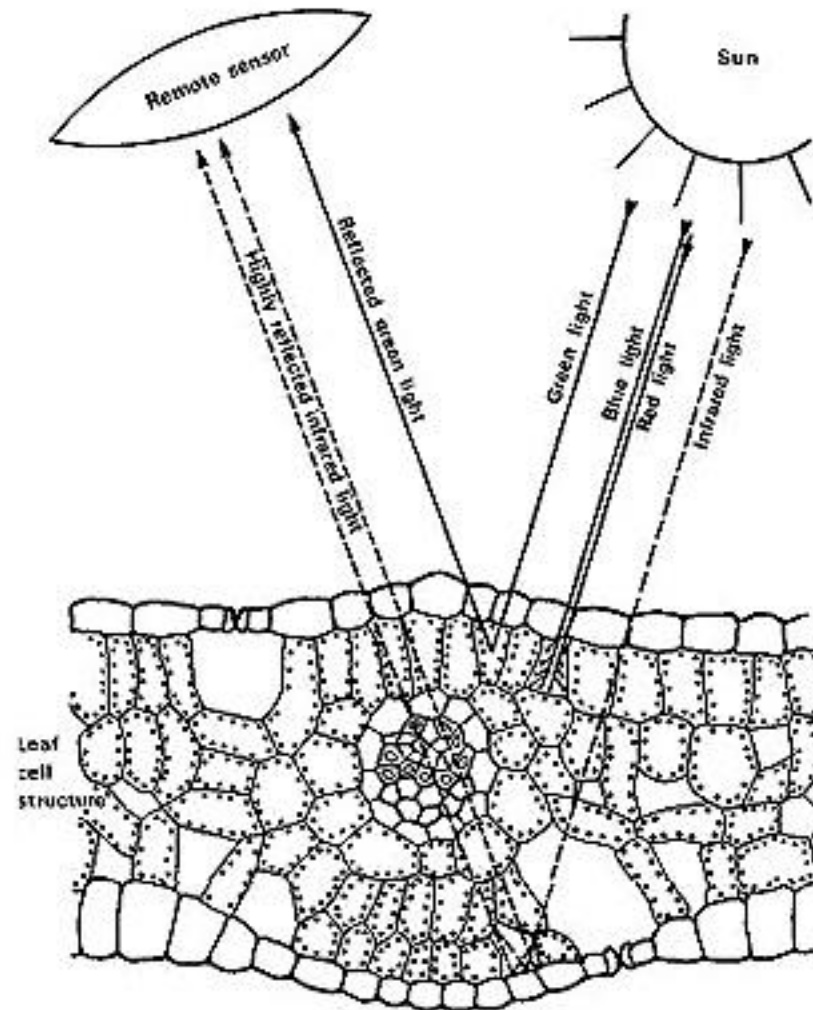


- Causes of spectral variation in reflectance?
 - (bio)chemical & structural properties
 - e.g. In vegetation, phytoplankton: chlorophyll concentration
 - soil - minerals/ water/ organic matter
- Can consider spectral properties as continuous
 - e.g. mapping leaf area index or canopy cover
- or discrete variable
 - e.g. spectrum representative of cover type (classification)

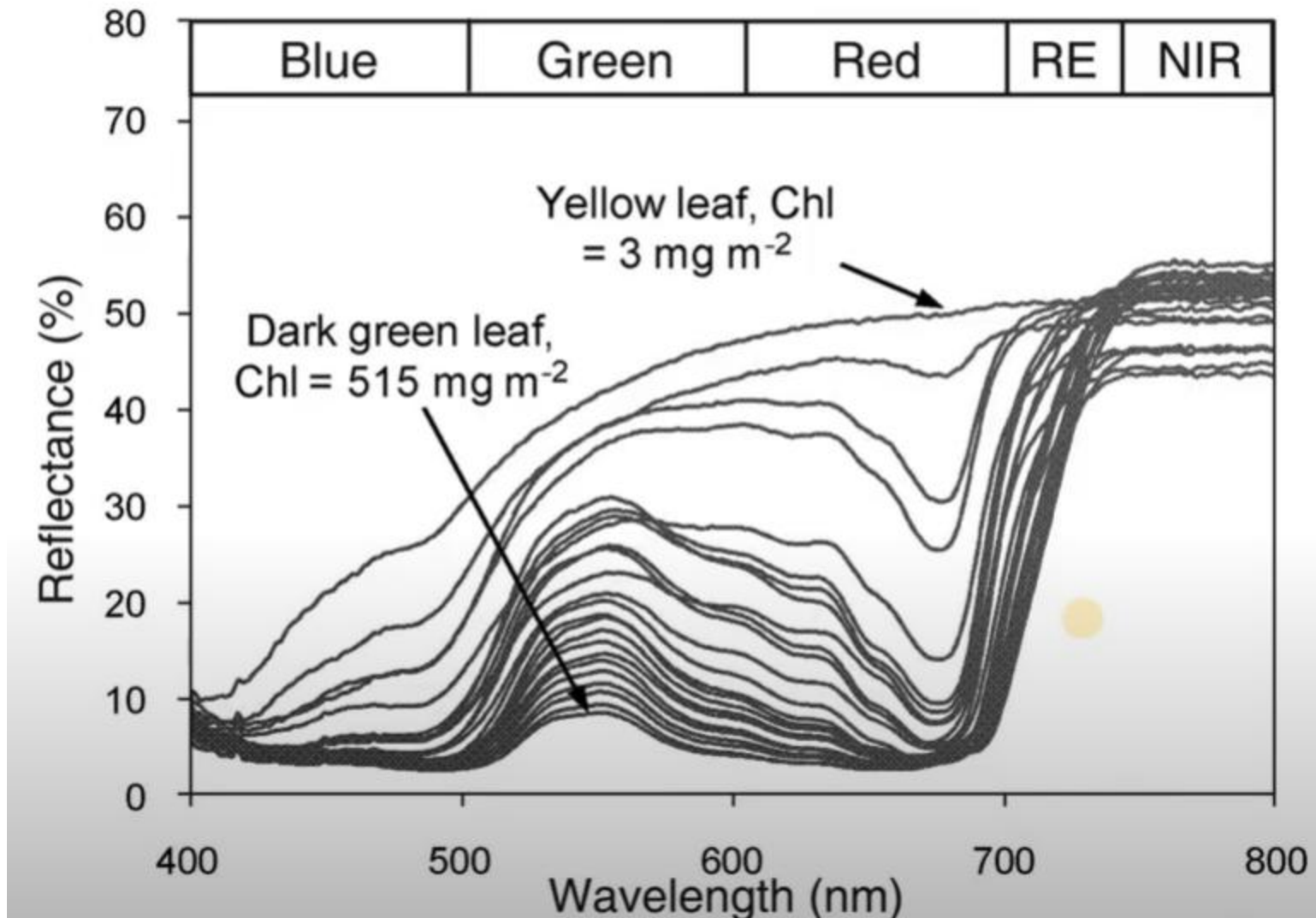
Surface spectral information: vegetation



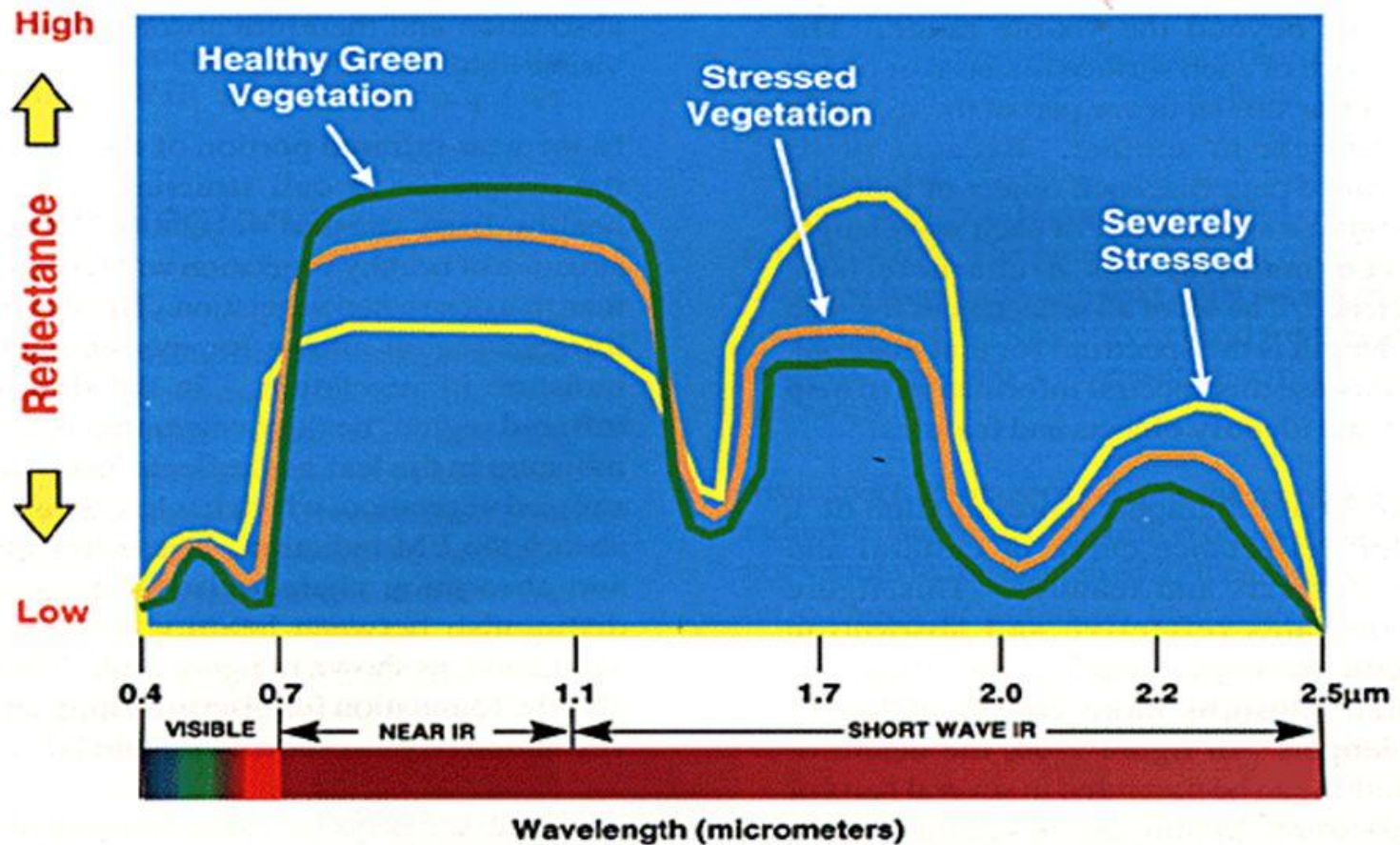
Surface spectral information: vegetation



Surface spectral information: vegetation



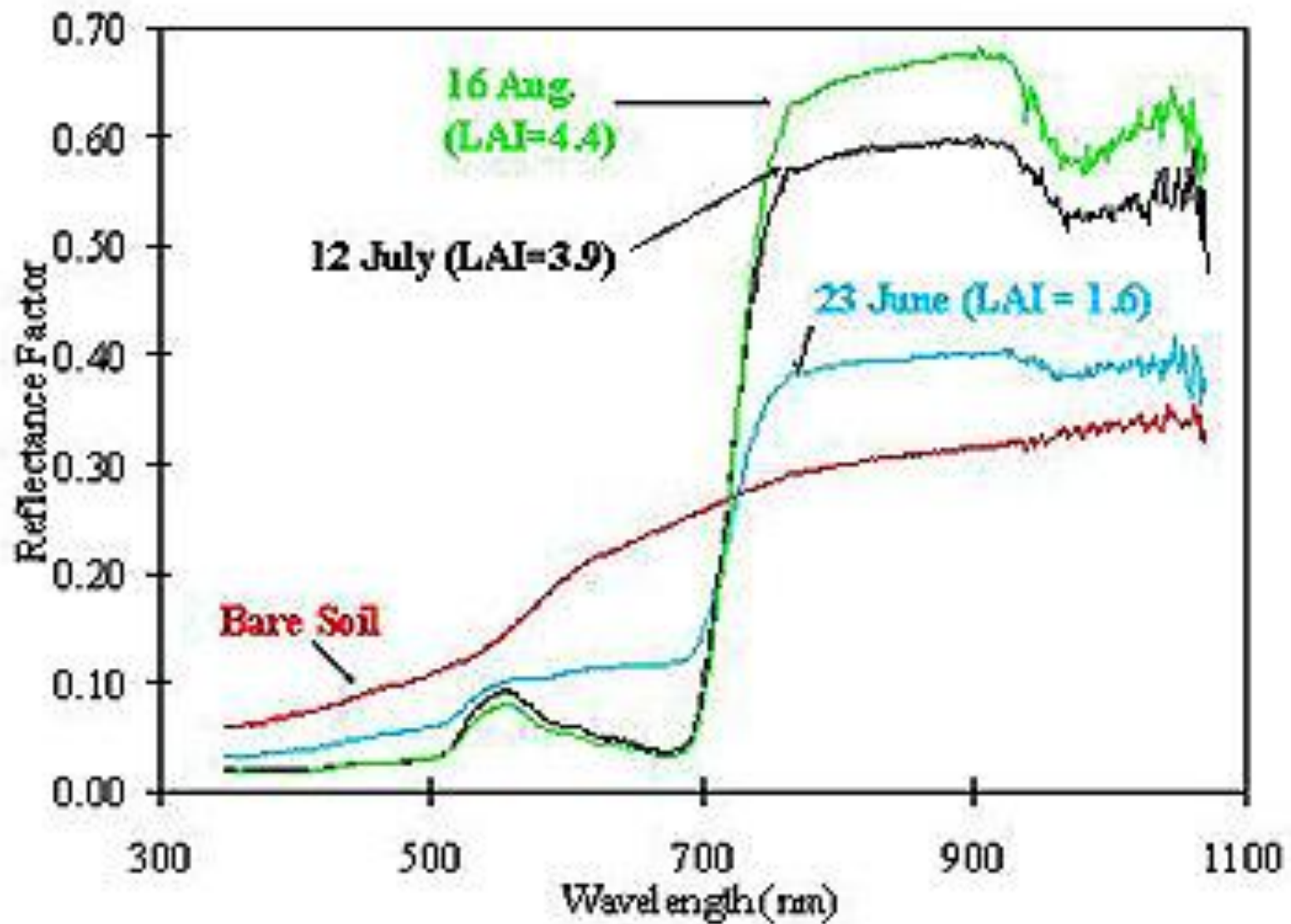
Surface spectral information: Vegetation



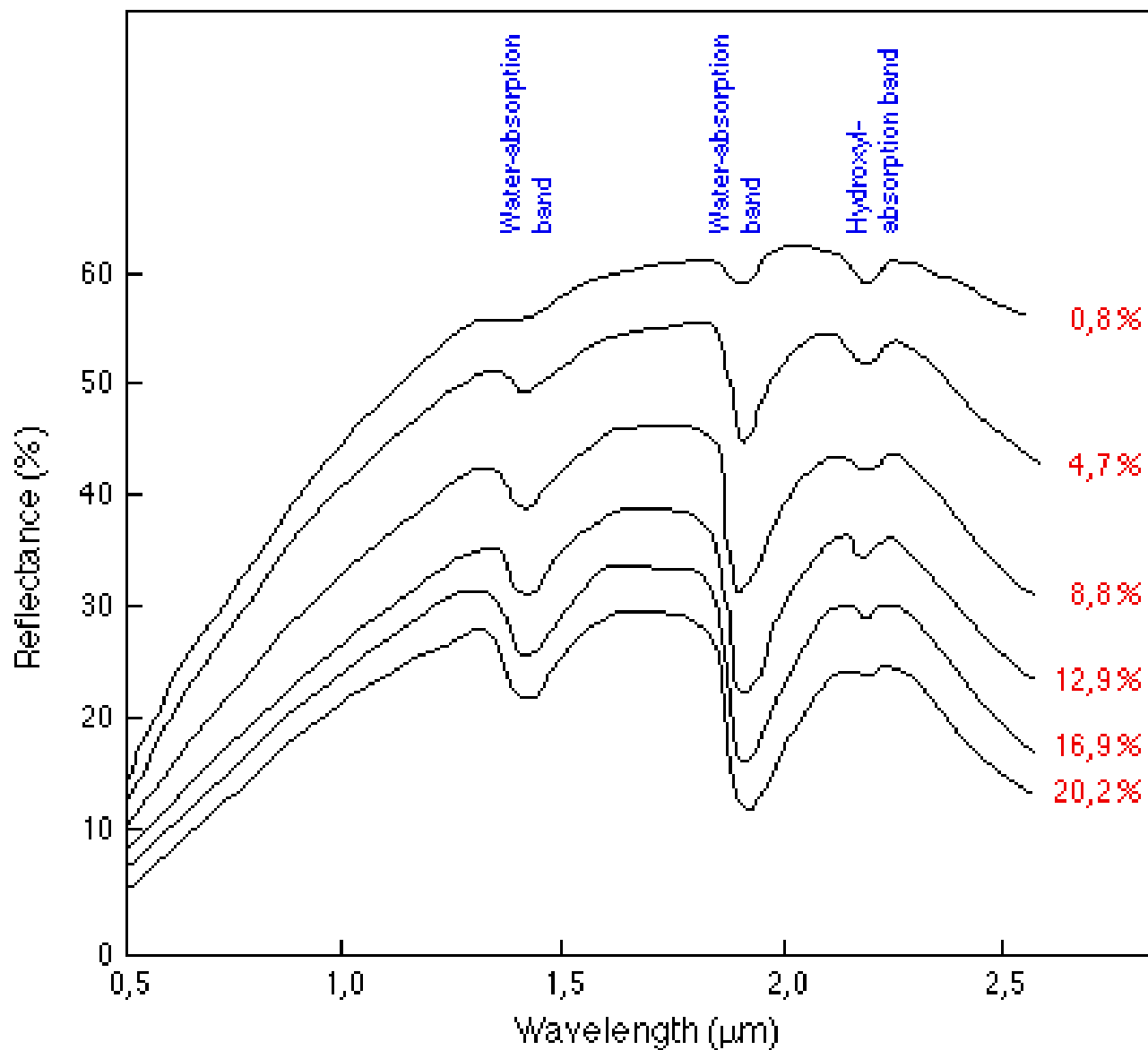
Use of Vegetation Spectral Signature

- forest inventories
- healthy vs. stressed vegetation

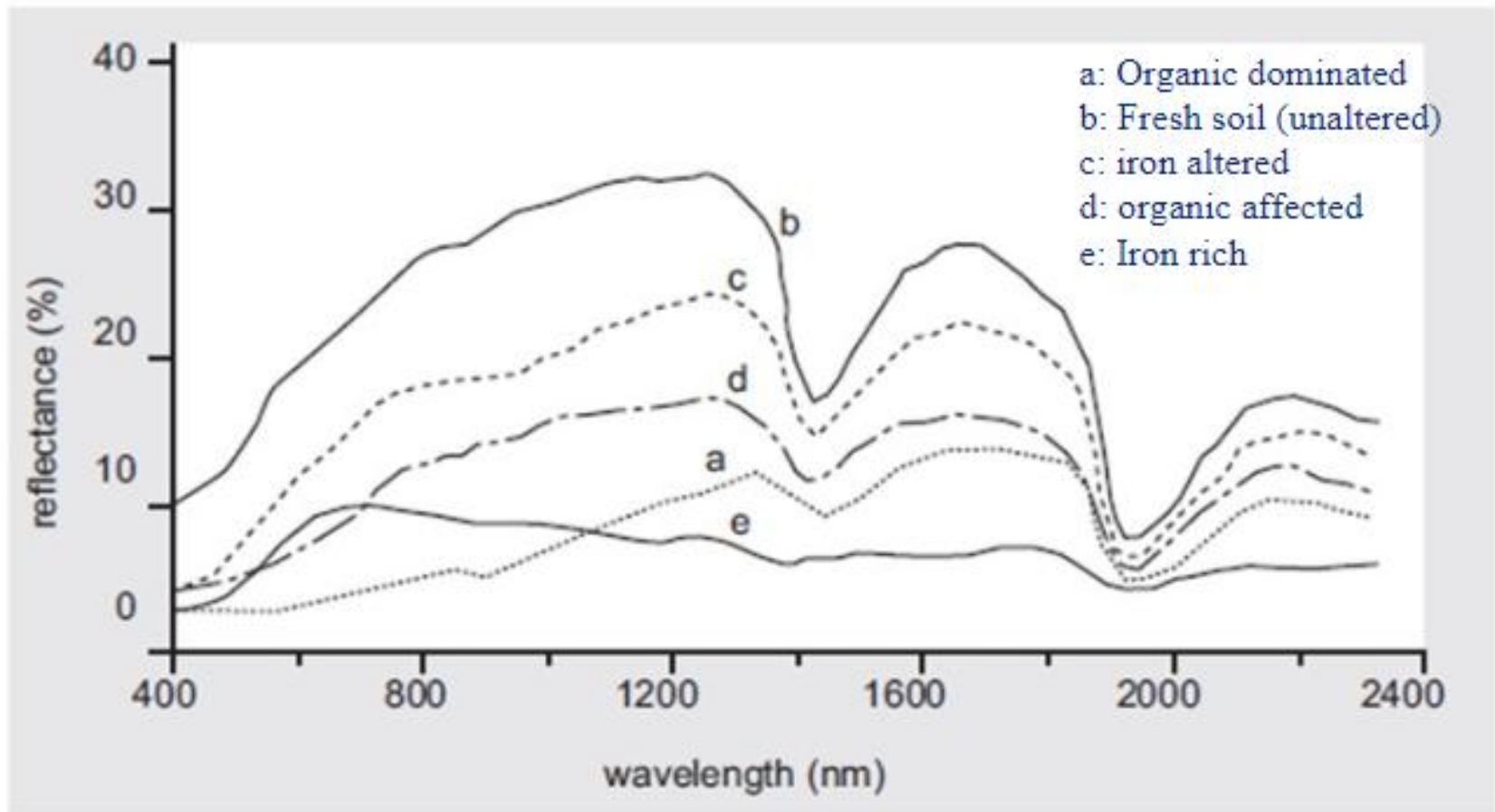
Surface spectral information: Vegetation



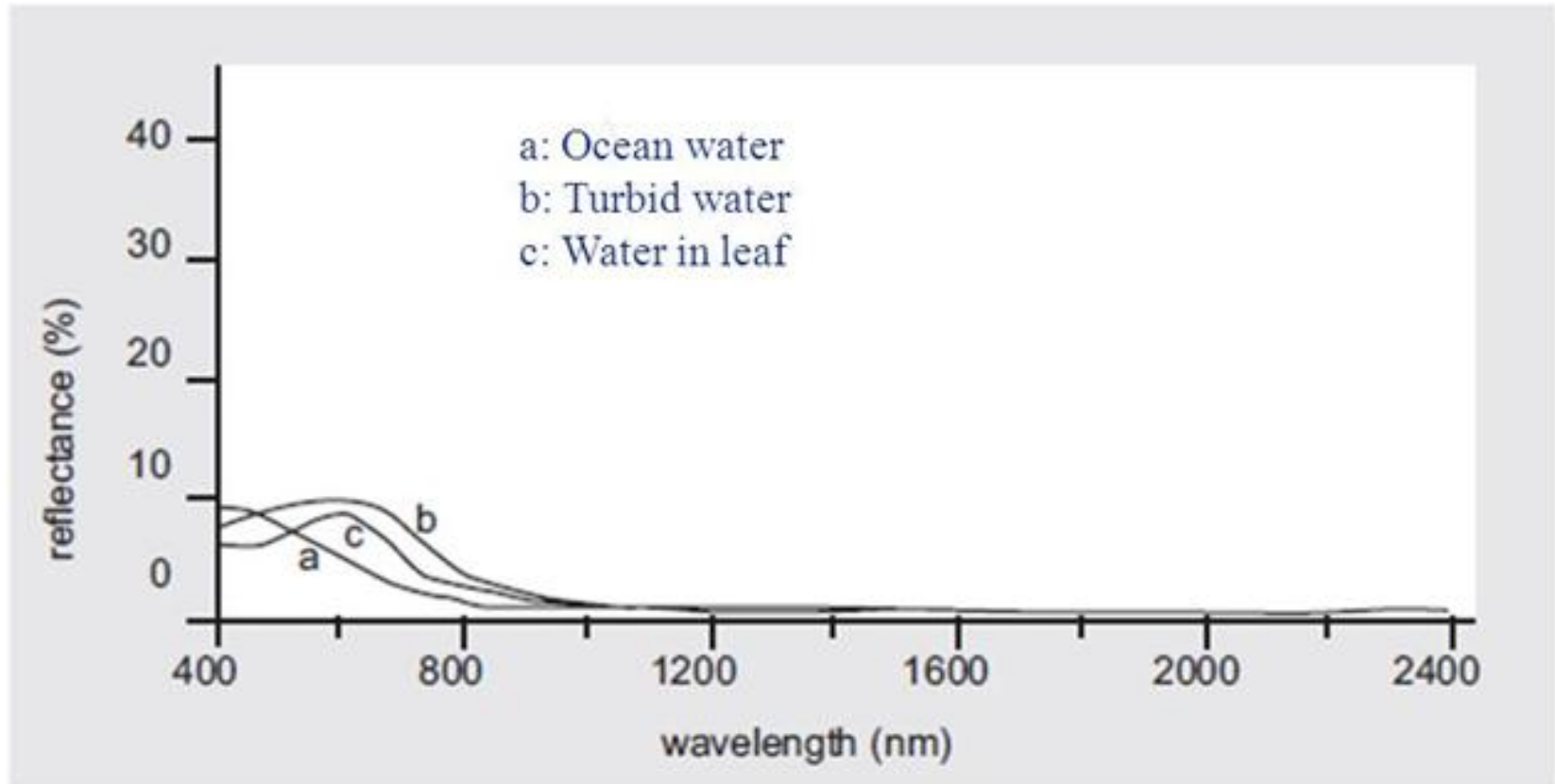
Surface spectral information: soil



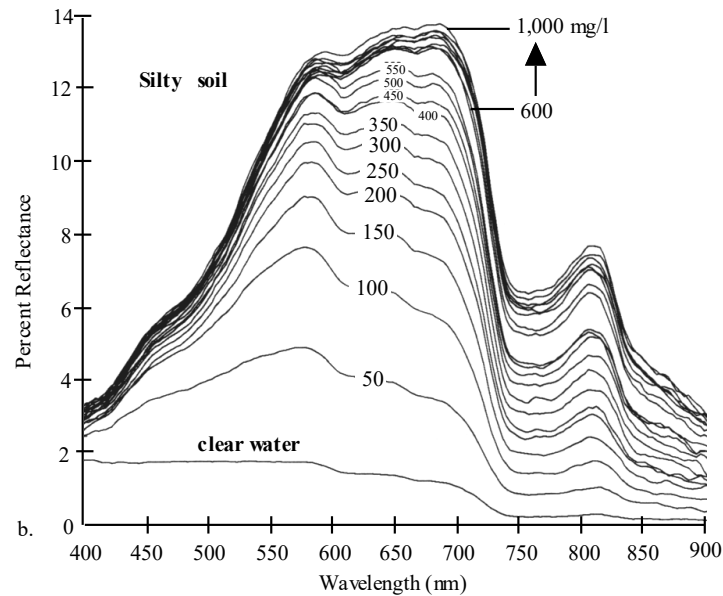
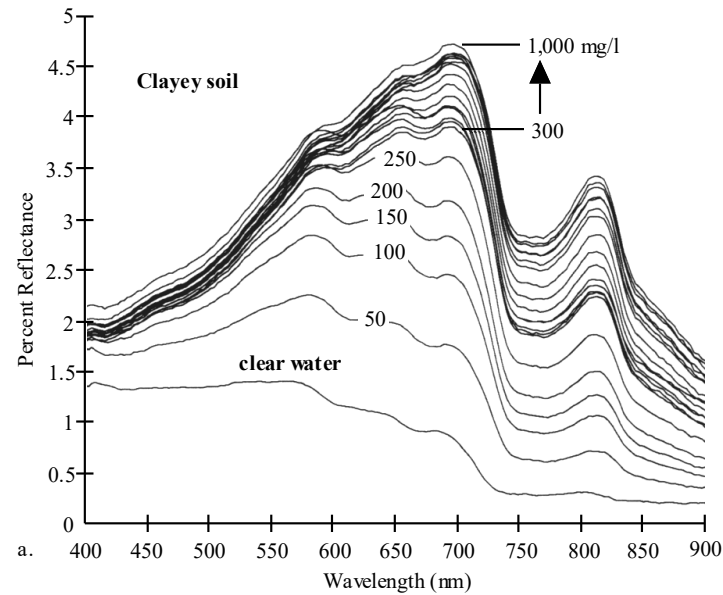
Surface spectral information: soil



Surface spectral information: Water

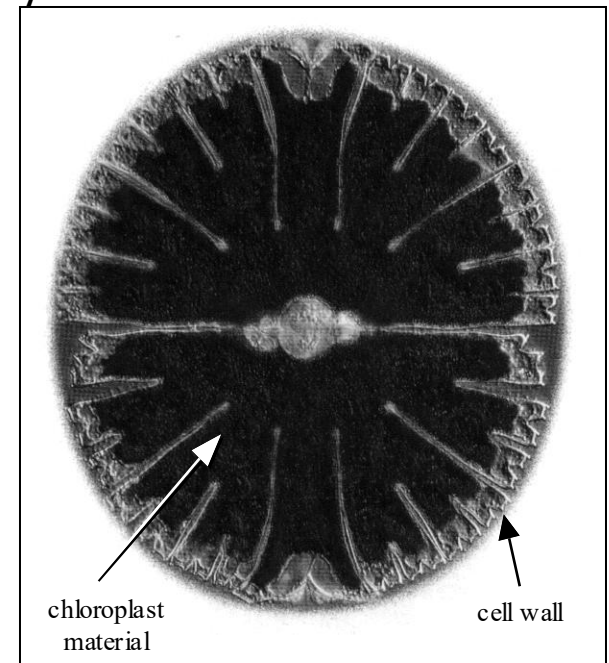
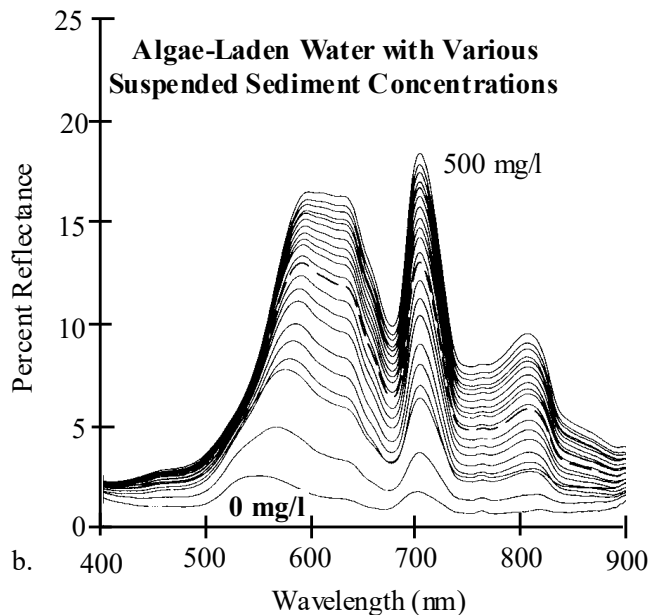
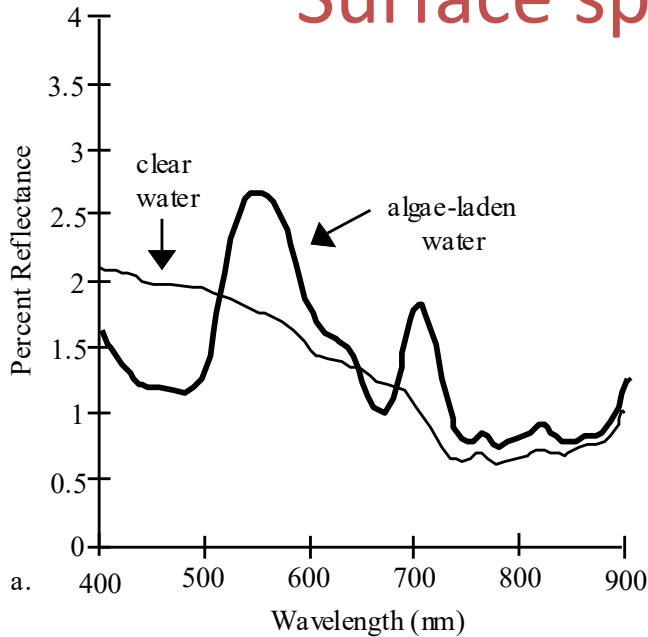


Surface spectral information: Water



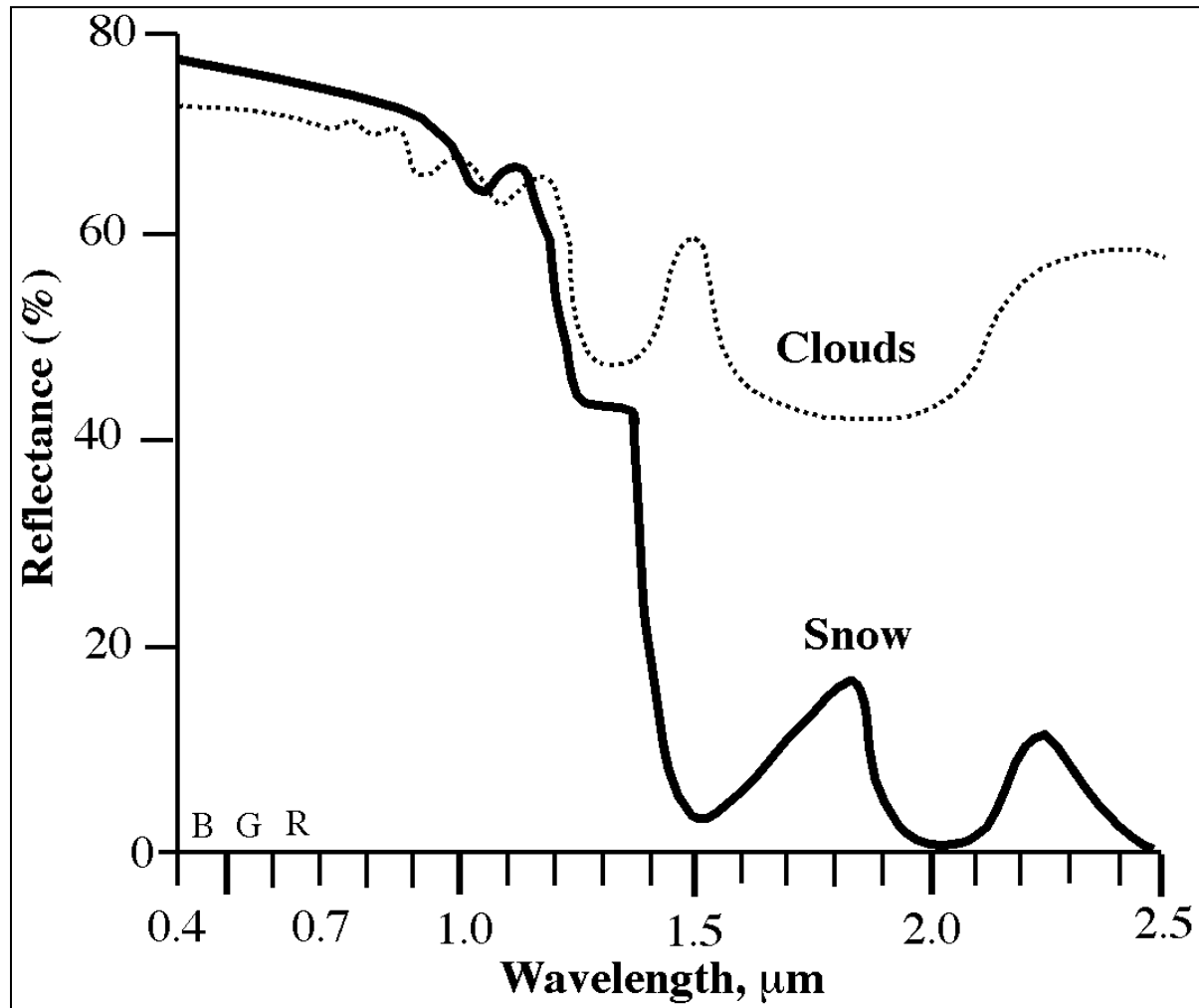
Surface spectral information: Water

Percent reflectance of clear and algae-laden water based on in situ spectroradiometer measurement. Note the strong chlorophyll a absorption of blue light between 400 and 500 nm and strong chlorophyll a absorption of red light at approximately 675 nm



Percent reflectance of algae-laden water with various concentrations of suspended sediment ranging from 0 - 500 mg/l

Surface spectral information: Clouds, Snow



2.4 INTERACTION AT THE SENSOR

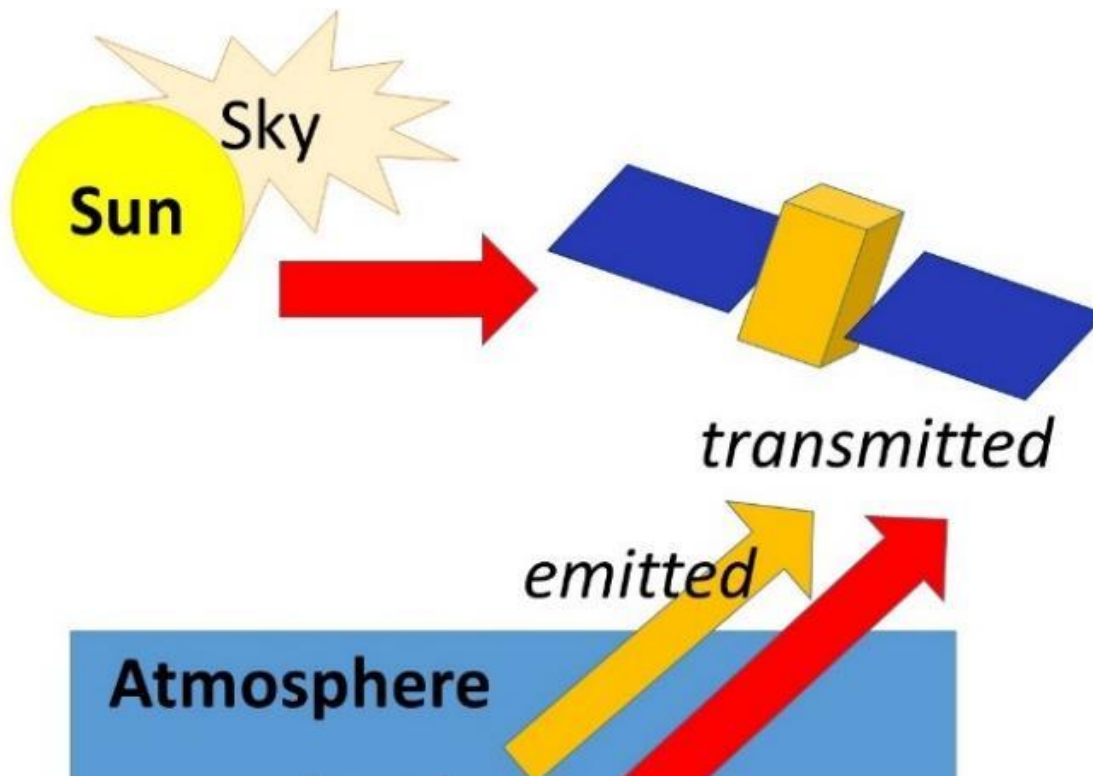
Extraterrestrial radiation:

Solar, Moon, star radiation
Galaxy
Cosmic background

Energy-matter

interactions:

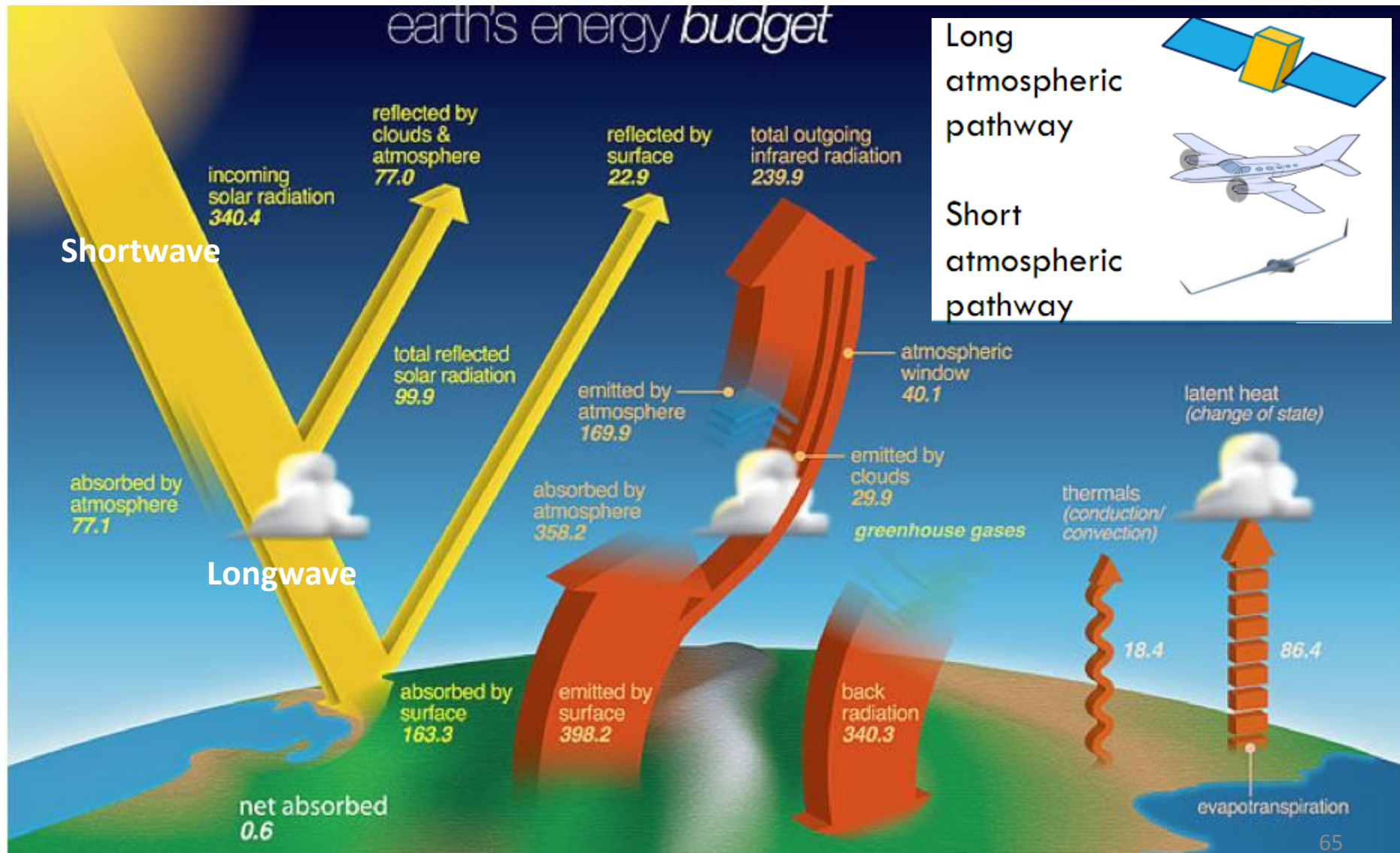
Camera film
Integrated circuits



>> Operation of sensors

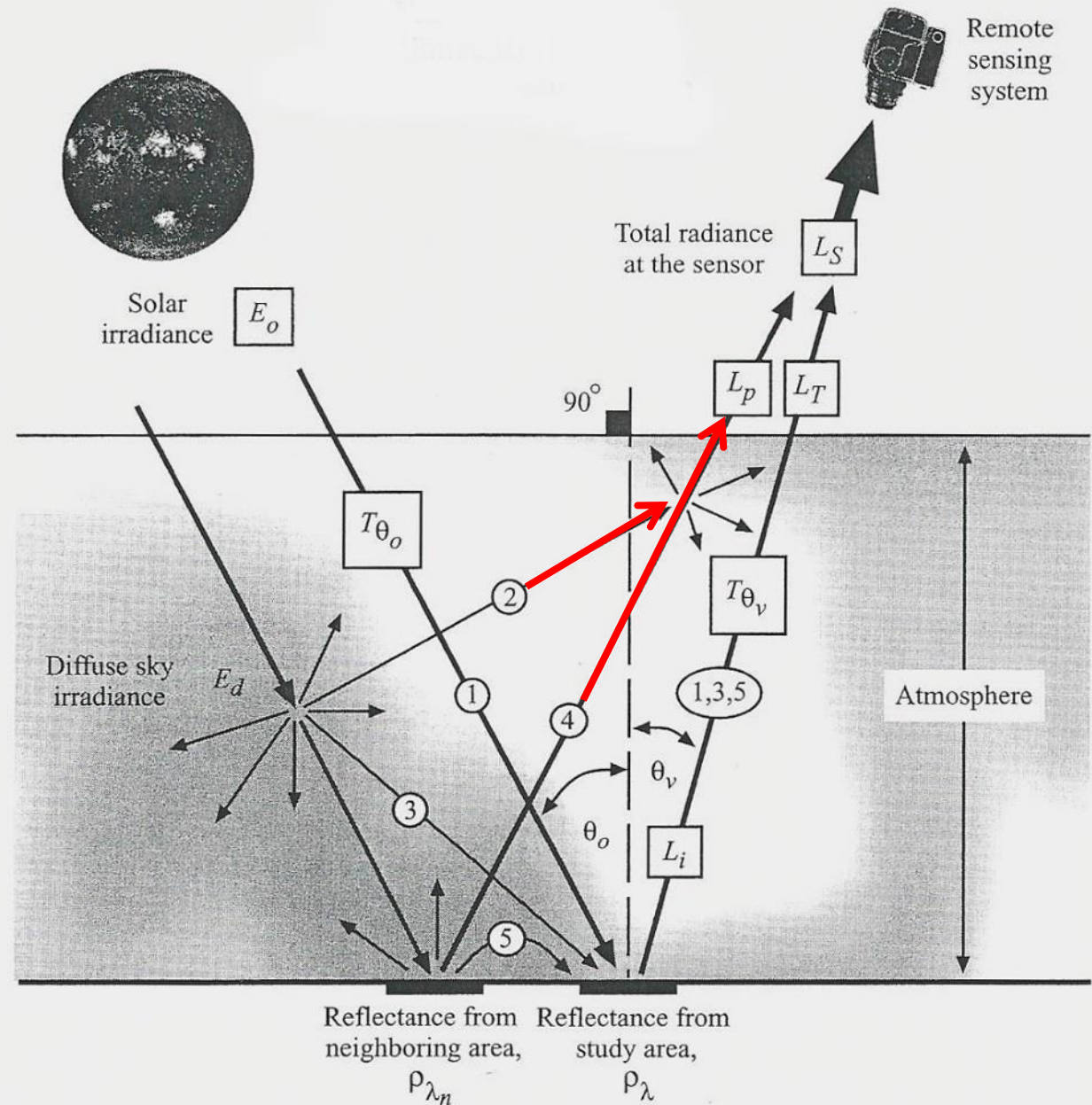
>> Sensor errors

RADIATION PATHWAYS



RADIATION PATHWAYS

unwanted
radiometric
effects/noise in
the remotely
sensed data



REMARKS

- **Electromagnetic Radiation:** essential for acquiring data about the Earth's surface

- Electromagnetic radiation **interacts with the atmosphere**

Important for **extract information** about characteristics of atmosphere
The interaction makes **noise** in surface images thus we need to do atmospheric correction for optical images.

- Electromagnetic radiation interacts with The Earth's Surface

Most important for acquire information of the surface objects
The **interaction varies** based on the different objects or conditions of objects on the surface.

- Electromagnetic radiation interacts at the Sensors

Sensor records reflectance radiation and convert to digital numbers
May make some noise in the image but the noises is **mostly systematically removed**.

Practice

- The Stefan-Boltzmann law : ➔ Estimate the total energy emitted by the Sun ?

$$R = \epsilon \sigma T^4$$

Radiation emitted

emissivity

Stefan-Boltzmann constant

absolute temperature

The diagram shows the equation $R = \epsilon \sigma T^4$. Red arrows point from each variable to its label: R to 'Radiation emitted', ϵ to 'emissivity', σ to 'Stefan-Boltzmann constant', and T^4 to 'absolute temperature'.

Stefan-Boltzmann constant: $\sigma = 5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$

Practice

- Planck's law is the general law for electromagnetic emission from the surface of a blackbody: ➔ Estimate the emission of Red wave length from the Sun?

$$W_{\lambda} = \frac{2\pi c^2 h \lambda^{-5}}{e^{\frac{ch}{\lambda KT}} - 1} [W m^{-2} \mu m^{-1}]$$

Symbol	Name	Value	Unit
c	Speed of Light	$2.99 \cdot 10^8$	$[m \cdot s^{-1}]$
h	Planck's constant	$6.62 \cdot 10^{-34}$	$[W \cdot s^2]$
K	Boltzmann constant	$1.38 \cdot 10^{-23}$	$[W \cdot s \cdot K^{-1}]$
T	Absolute temperature	≥ 0	$[K]$
W_{λ}	Radiant energy for unit area (emittance) at give λ		$[W \cdot m^{-2}]$
λ	Wavelength	≥ 0	$[\mu m]$

Thank you for your attention!